LSE CAPSTONE 2026

Beyond the Resource Curse

The Challenge of Industrial Upgrading in Resource-Rich Emerging Economies

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Chapter 1

Introduction

The endowment of natural resources formed the basis of early theoretical explanations for diverging growth trajectories across countries. Advancements in Chile and Malaysia demonstrate how natural resources can be leveraged successfully to vitalise a diversified economy. Yet these countries prove to be exceptions. Across much of the developing world, export baskets are characterised by persistent and heavy resource reliance.

The failure of resource wealth to deliver sustained growth has long occupied economists. The most influential framing, the resource curse, holds that resource endowments deterministically impede development. However, both the existence of this effect and the channels through which it operates have been subject to challenge. The literature has evolved from debating whether resources harm growth to identifying the conditions under which countries fail to convert extraction into broader productive capability. The answers point to interactions among financial depth, capital accumulation, institutional quality, and infrastructure. However, these structural factors are highly endogenous, rendering causal identification difficult. The econometric models employed in this report are grounded in intertemporal investment theory and carry a lagged structure, mitigating some sources of endogeneity. This approach is complemented by a model-agnostic analysis using machine learning techniques that identify relevant predictors beyond the theory-driven specification. Together, these analytical frameworks provide different perspectives that help us unravel the resource curse.

This report examines the extent to which resource-rich emerging economies diversify into higher value-added industries, using the Economic Complexity Index (ECI) to measure productive capabilities. The analysis combines a panel dataset of 54 resource-rich developing economies over 1995–2019. Finally, the case studies of the Republic of Congo, Azerbaijan, and Chile contextualise the cross-country findings.

Three pillars emerge as the most robust determinants of economic complexity: domestic credit to the private sector (financial depth), infrastructure capacity (electricity access), and human capital, both at a level and in interaction with resource intensity. Traditional macroeconomic variables contribute little predictive power once these structural determinants are accounted for. The case studies illustrate how binding constraints shift across settings: from infrastructure deficits and procyclical spending in Congo, to Dutch Disease and institutional weakness in Azerbaijan, to human capital and R&D gaps in Chile.

The content of this report proceeds as follows. Chapter 2 reviews the literature on resource curses. Chapter 3 presents the quantitative analysis, including econometric results, machine learning findings, and forward projections. Chapter 4 develops three case studies. Appendices provide technical and methodological detail.

Chapter 2

Literature Review

2.1 Resource Curse

The resource curse refers to the tendency of resource-rich countries to experience lower long-run economic growth despite their natural endowments. The Dutch Disease is a popular causal mechanism through which the resource curse operates. Resource-rent windfalls crowd out non-resource tradeable sectors. As countries discover and export subsoil assets, their currency appreciates, making domestic manufacturing uncompetitive in global markets (Corden and Neary, 1982).

One of the most influential early works on the resource curse used natural resource exports as a proxy of production, instrumenting with land area (Sachs and Warner, 1995). The study found that resource-intensive countries in 1971 exhibited lower growth rates between 1971 and 1989, relative to non-intensive ones. The literature further linked resource endowments to institutional deterioration, suggesting rent-seeking behaviour and authoritarianism (Ross, 2015). Collier and Hoeffler (2004) showed that commodity wealth increases the likelihood of rebellion and internal conflict.

However, in the early 2000s, Sachs and Warner (1995) faced criticism over endogeneity in their methodology, as natural resource exports themselves depend on private and public investment, institutional quality, access to electricity, and transport infrastructure. Since then, substantial research has found weak or no evidence for the resource curse (Poelhekke and van der Ploeg, 2010; Havránek, Horváth, and Zeynalov, 2016; Lashitew, Ross, and Werker, 2020; Lederman and Maloney, 2007). This shifted the literature towards identifying the specific conditions under which extraction leads to poor economic outcomes, rather than evaluating whether endowments are inherently detrimental.

2.2 Industrial Upgrading for Resource-Dependent Economies

2.2.1 The Logic of Industrial Upgrading

The modern understanding of economic growth rests on the premise that development is a process of accumulating productive capabilities and deploying them into increasingly complex activities. Hausmann, Hwang, and Rodrik (2007) show that what a country exports matters, as countries producing goods in higher value-added industries tend to grow faster.

This finding is formalised in the economic complexity framework of Hidalgo and Hausmann (2009), which maps the “product space” and demonstrates that countries diversify by leveraging existing capabilities to move into nearby, higher-value products. Industrial upgrading is not about leapfrogging into new sectors but about incremental “jumps” through the product space. This is based on accumulated know-how, measured by the Economic Complexity Index (ECI), which features in our empirical work.

The product space lays emphasis on the manufacturing sector as it is the most densely connected sector and generates the strongest learning-by-doing effects. When manufacturing contracts, a country loses not only its output but also the platform for future diversification. Resource-rich countries find their capabilities concentrated in extraction, far from higher-value manufactured goods in the product space. This “distance” in the network of the product space may further explain why some countries become trapped in resource extraction. The literature identifies several mechanisms through which this could occur.

2.2.2 Institutional Quality, Expropriation, and Rent-Seeking

A common argument is that institutional quality mediates the relationship between resource wealth and economic outcomes. Acemoglu, Johnson, and Robinson (2001, 2003) provide their theoretical foundation, showing that expropriation risk and property rights are the fundamental determinants of long-run development. Their framework implies that resource wealth interacts with inherited institutional structures rather than through independent effects on growth.

Mehlum, Moene, and Torvik (2006) further test this hypothesis by reproducing the Sachs and Warner (1995) regressions with an interaction term between resource abundance and institutional quality. They find that resources harm growth only where institutions are “grabber-friendly”: weak rule of law, poor contract enforcement, and high corruption. In “producer-friendly” institutional environments, resources are instead associated with faster growth.

This result has been influential in reframing the resource curse as conditional on governance. When institutions fail to constrain elite capture, resource rents are diverted from productive investment into patronage, consumption, and political consolidation. Acemoglu et al. (2019) further demonstrate that democratic institutions, which impose stronger constraints on rent extraction, are causally associated with higher long-run GDP.

2.2.3 Human Capital and Savings Rates

Resource dependence may also limit industrial upgrading through its effects on human capital accumulation and savings behaviour. Gylfason (2001) provides cross-country evidence that natural resource abundance is negatively associated with public education expenditure, school enrolment, and expected years of schooling. Resource booms raise the opportunity cost of education by offering high wages in extraction for unskilled labour, while governments with increased fiscal capacity face reduced incentives to invest in human capital, since their fiscal base does not depend on a skilled workforce.

Venables (2016) provides a comprehensive account of why resource-rich developing countries have struggled to convert subsoil assets into productive surface assets. He shows that adjusted net savings are strongly negative across low-income resource-rich economies, with governments consistently failing to channel rents into human and physical capital formation. Rather than investing in education, infrastructure, and institutional capacity, governments face intense pressure for current consumption, patronage spending, and inefficient transfers such as fuel subsidies.

These patterns reflect a deeper political economy problem. In resource-rich states, the social contract between taxation and public service provision is weakened because revenues derive primarily from extraction rather than from taxing a productive citizenry (Van der Ploeg and Venables, 2011; Venables, 2016). This generates underinvestment and decreases diversification incentives, even when formal institutional quality is not the binding constraint.

2.2.4 Price Volatility, Credit Ratings, and Investment

Primary commodity prices are inherently volatile, and this volatility transmits directly into the fiscal revenues, exchange rates, and growth trajectories of dependent economies. As such, they have gained prominence as a mechanism through which heavy primary resource dependence becomes costly.

Van der Ploeg and Poelhekke (2009) show that resource abundance boosts growth directly but undermines it through volatility, consistent with the conditional curse framework. Resources are not inherently harmful, but their macroeconomic side effects concentrate in countries lacking the institutional and financial capacity to manage them. Under credit constraints, volatility shifts investment composition away from long-term, high-return projects toward short-term, low-return alternatives (Aghion, Angeletos, Banerjee, and Manova, 2010).

These dynamics feed into sovereign creditworthiness. Mlachila and Ouedraogo (2020) find that commodity price shocks drive financial underdevelopment in resource-rich countries, mitigated only by strong governance. Manzano and Rigobon (2001) go further, arguing that the apparent resource curse might be a debt overhang problem: countries borrowed against future resource revenues during the 1970s commodity boom, and the resulting debt burden explains their subsequent poor growth. This interpretation reinforces the conditional view that the issue lies in the financial choices made because of resource abundance, rather than resource abundance itself.

2.2.5 Point-Source Resources as an Exacerbating Factor

Not all natural resources carry the same developmental risks. The literature distinguishes between geographically concentrated, capital-intensive commodities ("point-source") and more dispersed ones, produced by a broader base of economic actors ("diffused"). Isham et al. (2005) posit that point-source exports are associated with weaker institutional outcomes. This is because they generate larger, more concentrated rents that are easier for narrow elites to appropriate and harder for broad-based coalitions to monitor. Moreover, Boschini et al. (2007) show that the interaction between resource type and institutional quality is decisive: appropriable resources such as oil and diamonds are significantly more damaging in weak institutional settings but can be managed productively where institutions are strong.

These findings raise the question of where forestry sits within this typology. Conventionally grouped with agriculture as a diffuse, renewable resource, industrial forestry in many developing countries is governed more like mining. Tropical forests are overwhelmingly state-owned, with extraction rights allocated through concession contracts that mirror mineral concessions (Piketty et al., 2008; FAO and EFI, 2018). Corruption in these contracts parallels patterns in the mining sector, depriving governments of significant revenues through opaque allocation and regulatory capture (Søreide, 2007).

Despite these similarities, Bhattacharyya and Collier (2014) show that rents from non-renewable mineral depletion reduce the public capital stock, while rents from forestry and agriculture do not. Thus, forestry occupies an intermediate position between the point-source and diffuse categories.

Chapter 3

Quantitative Analysis

The literature review establishes two broad findings. First, the negative causal relationship between resource abundance and economic growth lacks consistent empirical support. Second, that resource-rich countries' development outcomes depend on interacting factors, including institutional quality, human capital, fiscal management, infrastructure, and resource type.

Our objective is to assess the ability of resource-rich countries to upgrade their value chain, the process by which economies move from exporting commodities to more knowledge-intensive activities. To do this, we employ the Economic Complexity Index (ECI) as the outcome of interest, which measures the diversity and sophistication of a country's export basket (see Appendix A). Our quantitative models evaluate the relative importance of the factors identified in the literature and their relationships with natural resource production.

To conduct the quantitative analysis, two complementary empirical approaches are employed. The econometric framework uses panel regressions to test a specific hypothesis drawn from the theoretical discussion. Based on the intertemporal investment argument of Venables (2016), human capital and physical capital accumulation interact with the intensity of resource production to predict changes in economic complexity. This allows us to estimate interpretable coefficients and compare our results directly with prior work. The machine learning (ML) framework takes a more agnostic approach. By including a broader set of variables and allowing the data to determine which predictors matter most, the ML analysis can detect patterns that the theory-driven econometric model may miss, while also cross-validating the econometric findings.

3.1 Data

We construct a country-level panel dataset covering the period 1995–2019. This incorporates variables across a broad set of dimensions that may influence an economy's capacity to move into more complex and higher value-added activities. The dataset includes: (i) natural resource endowment indicators, covering both resource rents and production volumes; (ii) macroeconomic variables; (iii) fiscal and financial indicators; (iv) governance measures; (v) indicators capturing the GDP structure of each country; (vi) human capital variables; (vii) infrastructure indicators; and (viii) geographical and other structural characteristics.

The variable choices and their sources are summarised in Table 3.1 and Figure 3.1. Most of the data draws from the World Bank's World Development Indicators, which supplies most macroeconomic, structural, financial and human capital variables. Governance indicators are sourced from the Varieties of Democracy (V-Dem) dataset. Factor inputs and productivity measures come from the Penn World Tables version 11.0 (PWT 11.0). Macroeconomic flow variables, including government finance aggregates, are supplemented from the International Monetary Fund (IMF). Natural resource production data are drawn from the Energy Institute and Our World in Data (EI/OWID). Additional details on each variable and its source are provided in Appendix A.

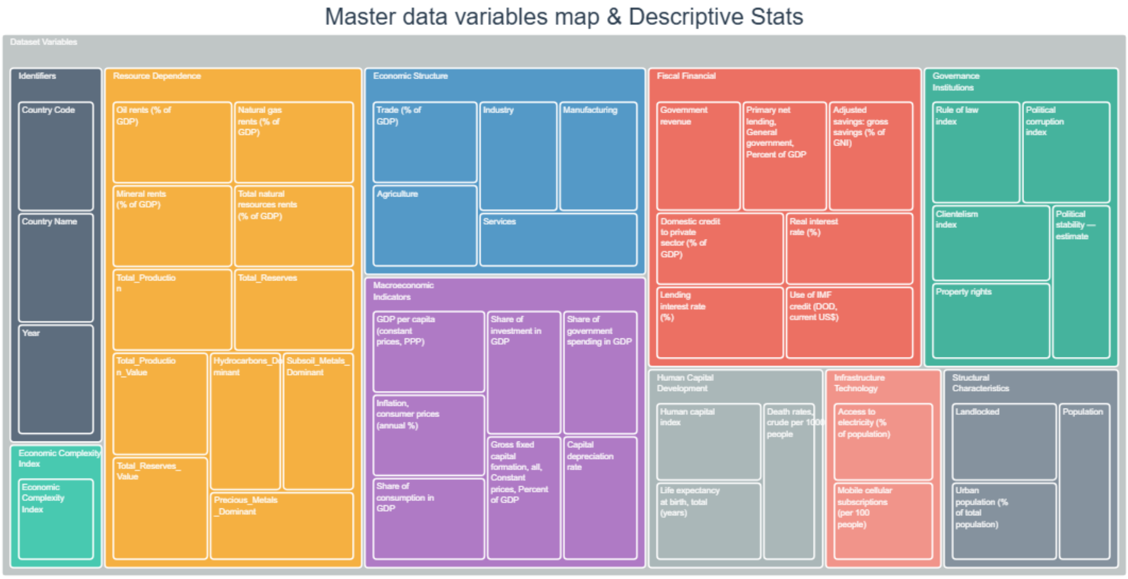

Table 3.1. Data variables

Figure 3.1. Data sources by macro-indicator

3.1.1 Country Selection

To avoid arbitrarily selecting our countries of interest, we conduct our analysis by excluding countries classified as high income by the World Bank where natural resource rents are lower than 5% of GDP in 1995. This threshold is intended to ensure that natural resources represent a significant component of economic activity. A special treatment is given for Gulf states, as their 1995 income class does not reflect the industrial upgrading to the same degree as, for instance, European economies. Therefore, despite being classified as high-income in 1995, these countries are retained in our sample.

Several countries that satisfy these criteria are excluded on data availability grounds. The complete list and the rationale for each exclusion are provided in Appendix B. The final sample comprises 54 countries, shown in Figure 3.2. Any remaining gaps are filled by linear interpolation within each country's time series, with extrapolation capped at three years beyond the nearest observed value to avoid projecting short-run patterns over long horizons (Appendix B). Where a country has no observed value for a variable across the entire sample period, a second pass uses K-Nearest Neighbours (KNN, $k = 5$, distance-weighted) to fill the missing cell from the five most similar country-year observations (Appendix B).

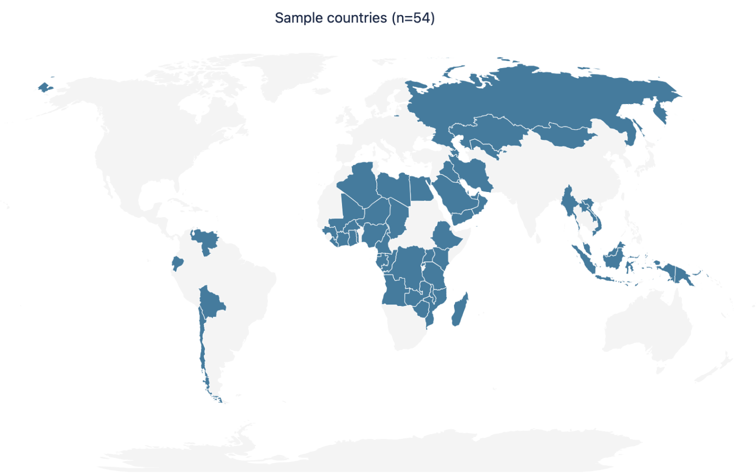

Figure 3.2. Countries in the sample

3.2 Descriptive Statistics of the Sample

The high-resource subsample shows a lower mean ECI (-0.760 versus -0.070), consistent with resource-rich economies having a generally lower level of productive diversification. Human capital, proxied here by the Human Capital Index (HCI), is also modestly lower in the high-resource subsample, while natural resource production value per capita is, by construction, substantially higher.

These patterns reflect well-documented features of resource-dependent economies. Productive diversification into more complex goods requires functioning credit markets, a skilled workforce, and the institutional conditions that correlate with lower corruption and longer life expectancy. Table 3.2 presents descriptive statistics for the full sample of 3,150 country-year observations and for the restricted subsample of high-resource countries (1,350 observations). Figure 3.3 shows Pearson correlations between ECI and the candidate variables. The strongest positive associations are with domestic credit, the human capital index, and life expectancy. The strongest negative is with political corruption and agricultural value added as a share of GDP, the latter reflecting the standard structural shift away from primary activity as complexity rises. These correlations motivate the variable selection in the regression and machine learning stages that follow.

Descriptive Statistics

	Full Sample								High-Resource Countries							
	N	Mean	Std. Dev.	Min	p25	Median	p75	Max	N	Mean	Std. Dev.	Min	p25	Median	p75	Max
Continuous variables																
Economic Complexity Index (ECI)	3,150	-0.070	1.024	-2.890	-0.832	-0.209	0.586	2.718	1,350	-0.760	0.670	-2.890	-1.220	-0.804	-0.343	1.211
Human Capital Index	3,150	2.457	0.690	1.049	1.876	2.504	3.019	3.813	1,350	2.062	0.559	1.049	1.631	2.068	2.463	3.471
Gross Fixed Capital Formation (% of GDP)	3,150	18.207	8.223	-0.000	12.544	18.401	23.224	67.196	1,350	16.739	9.431	-0.000	10.008	14.959	23.020	67.196
Political Stability	3,150	-0.209	0.947	-3.180	-0.853	-0.183	0.522	1.759	1,350	-0.567	0.850	-3.180	-1.165	-0.516	0.067	1.224
Rule of Law Index	3,150	0.540	0.309	0.011	0.264	0.530	0.856	0.999	1,350	0.359	0.225	0.011	0.173	0.327	0.545	0.961
Natural Resource Production Value Per Capita	3,150	908.631	2851.017	0.000	13.707	95.495	388.730	40646.664	1,350	1526.635	3834.257	0.000	21.178	196.958	1221.262	40646.664
Trade (% of GDP)	3,150	74.311	34.389	0.021	48.810	67.531	91.695	257.653	1,350	72.319	33.351	0.021	47.835	63.765	90.021	220.407
Access to Electricity (% of pop.)	3,150	77.159	31.760	1.028	55.925	97.900	100.000	100.000	1,350	60.126	36.696	1.028	22.450	64.800	99.000	100.000
Gross Savings (% of GNI)	3,150	23.727	11.854	-29.263	16.335	22.538	30.298	68.544	1,350	25.020	15.010	-29.263	15.228	24.657	33.700	68.544
Domestic Credit to Private Sector (% of GDP)	3,150	48.751	44.422	0.491	15.695	33.151	69.088	254.668	1,350	25.492	25.319	0.491	9.201	16.795	32.869	158.505
Binary variables (share = mean)																
Hydrocarbons Dominant (=1)	3,150	53.7%	—	—	—	—	—	—	1,350	56.1%	—	—	—	—	—	—
Subsoil Metals Dominant (=1)	3,150	23.5%	—	—	—	—	—	—	1,350	14.8%	—	—	—	—	—	—
Precious Metals Dominant (=1)	3,150	19.3%	—	—	—	—	—	—	1,350	26.0%	—	—	—	—	—	—
Observations	3,150								1,350							
Countries	—								—							

Note: Statistics computed on non-missing observations. Binary variables report the share of country-year observations equal to 1. Std. Dev., Min, quartiles and Max are omitted for binary variables.

Table 3.2. Descriptive statistics

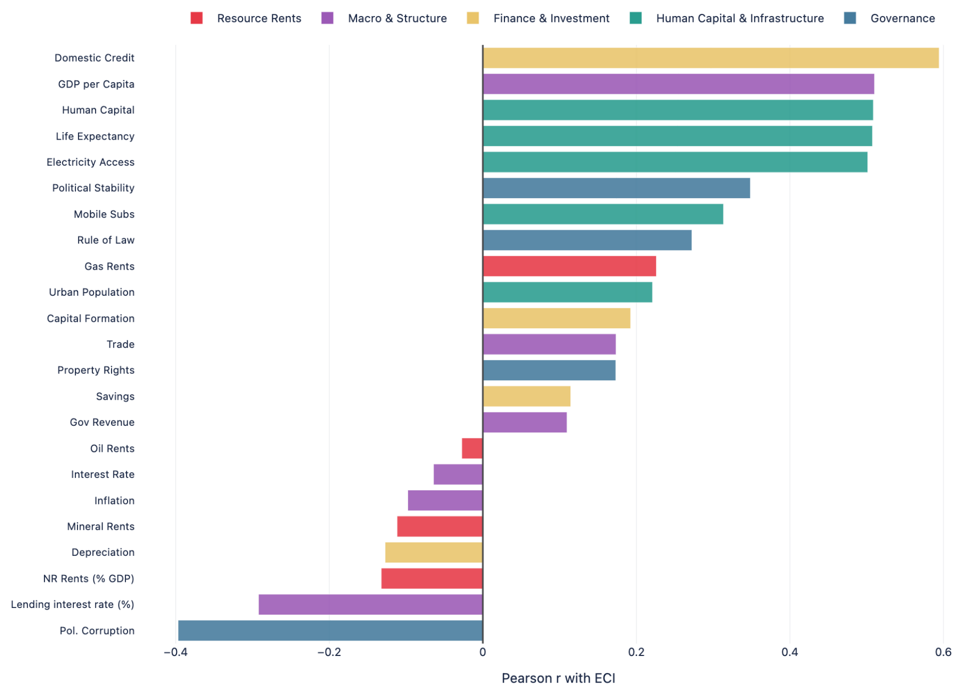

Figure 3.3. Correlation of features with ECI

3.3 Clustering Countries by Natural Resource Production

As an exploratory exercise, we conduct clustering, a technique that identifies groups of similar observations based on their characteristics. In our case, clusters are based on the resource profile of each country, grouping them in different sections. This helps us understand the various types of resource intensiveness.

The clustering proceeds in two stages. First, we apply Principal Component Analysis (PCA), which compresses many correlated resource variables into just two, while still capturing most of the variation in the data. Second, we apply K-means clustering to these summary dimensions, where each country is assigned to a group based on how similar it is to other countries. A more in-depth explanation of the clustering process is included in Appendix C.

3.3.1 Principal Component Analysis

PCA transforms the production values for each resource of interest into a smaller set of variables. These are called principal components (PCs), each constructed to capture as much of the remaining variance in the data as possible. Compressing multiple indicators into just a few discards some information about the sample, but by maximising the variance retained at each step, PCA limits this loss while improving interpretability.

We carry out this technique for two reasons. First, rather than preselecting which resources to cluster on, the approach allows the data itself to reveal which minerals and hydrocarbons account for the most cross-country variation. Second, it provides a less noisy set of inputs to the K-means algorithm, which tends to improve both clustering stability and interpretability.

We retain the first two principal components, which together account for 70% of the total variance in the standardised data. Around 30% of the information available in the sample is lost, but this is an acceptable trade-off given the reduction from 21 resource variables to just two composite dimensions. PC1 captures the largest share of variance and loads most heavily on oil and natural gas, effectively separating hydrocarbon producers from the rest of the sample. PC2 is driven by copper, gold, and coal, distinguishing countries whose extraction is concentrated in mining. The full set of factor loadings and variance shares are reported in Figure 3.4 and discussed in Appendix C.

3.3.2 K-means Clusters

To group countries with similar extraction profiles, we apply K-means clustering to the country value on the first two principal components. K-means partitions observations into a pre-specified number of groups by minimising the distance (extraction profiles) between members of the same group (or cluster). We set the number of clusters to five and use production data for 1995; Appendix C discusses choices regarding the number of clusters, the year selected, and the variables included. The resulting groupings are described below:

1. **Petrostates** comprise countries with very high oil or gas endowments and limited mineral production. Saudi Arabia, Kuwait, the UAE, Qatar, Oman, and Bahrain all mainly rely on hydrocarbons and have high production figures per capita.
2. **Oil Exporters** captures countries with moderate oil or gas output alongside limited mineral activity. Unlike the petrostate group, these economies are less dominated by a single commodity and tend to have lower per-capita hydrocarbon revenues. The cluster includes Ecuador, Angola, Nigeria, Egypt, Iraq, Libya, Algeria, the Republic of Congo, Azerbaijan, Trinidad and Tobago, and Gabon.
3. **Major Producers** groups countries with significant and often diversified production spanning both hydrocarbons and minerals. Russia, Kazakhstan, Indonesia, Papua New Guinea, Uzbekistan, Bolivia, and Malaysia occupy a wide range of extraction profiles.
4. **Mining Exporters** contains Chile and Mongolia, which produce mainly minerals rather than hydrocarbons.
5. **Forestry Intensive** includes countries with smaller-scale or less diversified extractive sectors, such as the Democratic Republic of Congo, Zimbabwe, Ghana, Guinea, Mali, Niger, and Vietnam. Many of these countries entered the sample primarily through forestry rents rather than mineral or hydrocarbon extraction.

Figures 3.4 and 3.5 confirm that these five groups occupy distinct regions of the resource space. In the PCA biplot (Figure 3.4), Petrostates and Oil Exporters are pulled toward the right by oil and gas loadings, Mining Exporters sit high on the copper and gold axis, and Forestry Intensive countries cluster in the lower-left corner with weak loadings on all major resources. The clusters are not regionally determined, as they often span continents. Oil Exporters are present in West Africa, North Africa, and the Caribbean, while Major Producers stretch from Central Asia to Southeast Asia and Latin America (Figure 3.5).

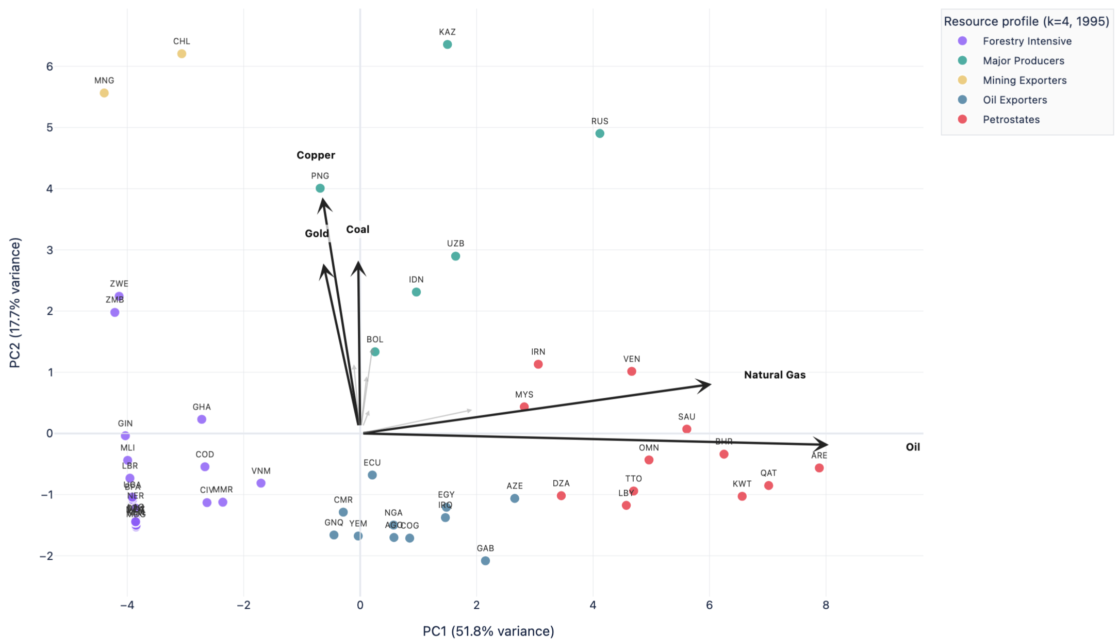

Figure 3.4. K-means clusters in PCA space ($K = 5$)

Understanding how these groups differ in ECI helps identify how each resource type correlates with development. Figure 3.6 plots the median ECI for each cluster between 1995 and 2019. Petrostates remain the most complex economies in the sample, while Oil Exporters consistently record the lowest median complexity. Resource type alone does not explain these gaps: the two hydrocarbon-based clusters end up at opposite ends of the complexity distribution. Oil Exporters record median ECI levels around 0.2 to 0.3 points below Forestry Intensive economies throughout the panel (Figure 3.6), despite recording

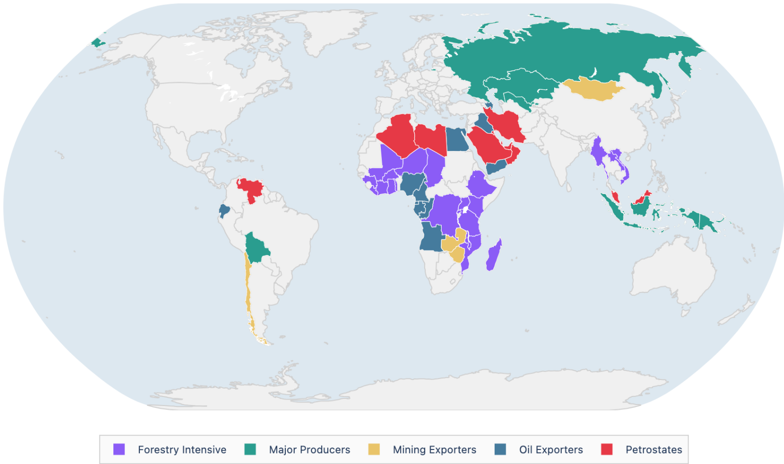

Figure 3.5. Natural resource clusters mapped by country (1995)

the highest normalised oil rents in the sample (Figure 3.7). This aligns with the point-source versus diffuse resource distinction: hydrocarbon rents are easier for narrow elites to capture, while forestry operates through a broader economic base. Major Producers and Mining Exporters converge near the middle of the distribution by 2019.

Figure 3.7 completes the picture. Petrostates approach Mining Exporters in GDP per capita, while Oil Exporters and Forestry Intensive countries rank lowest on income, credit access, and human capital. Mining Exporters, driven largely by Chile, record the strongest governance and human capital scores. Governance quality, human capital, and credit access vary widely even within similar resource bases, suggesting these characteristics matter more for complexity than the commodities extracted. The econometric and machine learning analyses that follow test this formally.

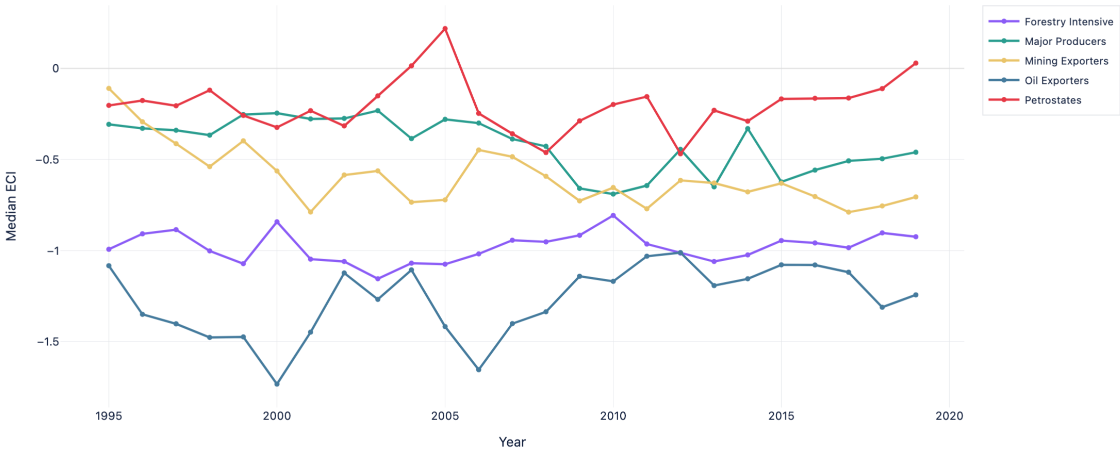

Figure 3.6. Evolution of Economic Complexity vs. log GDP per capita (PPP), 1995–2019

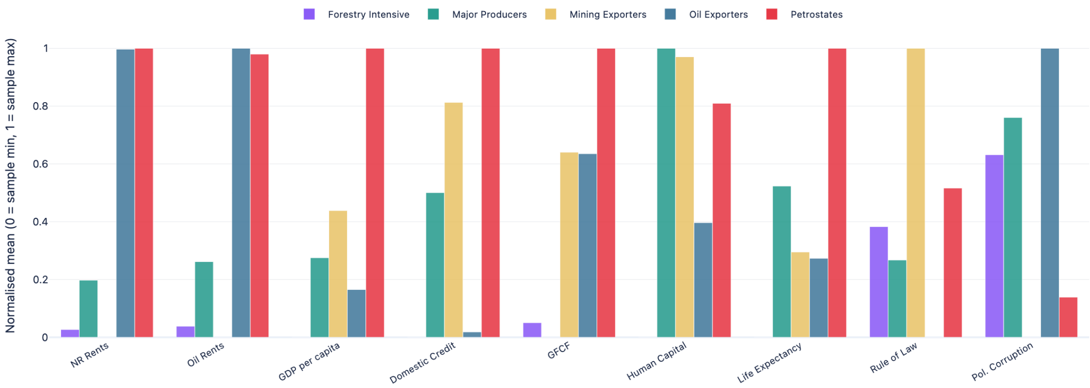

Figure 3.7. Difference between clusters

3.4 Econometrics Approach

3.4.1 Background

The econometric analysis tests a conditional resource explanatory mechanism. While most empirical research has focused on institutional quality as the primary mediating factor (Acemoglu, Johnson, and Robinson, 2001; Mehlum, Moene, and Torvik, 2006), the role of how resource revenues are saved and invested has received less attention. Therefore, we rely on Venables' (2016) intertemporal allocation framework, where resource-rich countries choose between consuming extraction revenues or converting them into physical infrastructure, human capital, and financial assets that sustain income post-depletion. While poorer countries may save less given higher marginal utility of consumption, Venables argues that observed savings rates in resource-rich economies remain below optimal levels, constraining structural transformation.

We test this argument empirically by testing whether capital accumulation, both physical and human, is associated with higher economic complexity, and whether this association is stronger in countries with more intensive resource production. If the framework is correct, the investment returns are greater where resource production is large, because in those settings the gap between current and optimal capital stocks is widest and the potential gains from investment are greatest.

We also test for differentiated impacts from subsoil versus forest assets. Given that subsoil resources (oil, gas, coal, minerals) are more geographically concentrated, point-source commodities would make this effect more pronounced, yielding larger investment returns than for forest assets.

3.4.2 Model

Estimating the impact of capital accumulation on economic complexity in resource-dependent countries is challenging due to endogeneity. Discovering and extracting resources require a productive capability baseline. However, causation among institutions, investment, and extraction is difficult to establish. Good institutions may give rise to the firms and coordination needed to operate mining activities, or alternatively, income from extraction may fund the institutional development that producers demand to protect their returns.

Similar arguments apply to trade openness, credit access and infrastructure: each plausibly causes and is caused by the level of economic complexity. We cannot fully resolve these issues and therefore do not infer causality.

The Human Capital Index (HCI) and Gross Fixed Capital Formation as a share of GDP (GFCF) serve as measures of human and physical capital accumulation, respectively. We use production to capture natural resource intensity measured as the total value of natural resource assets produced each year and calculated as physical output quantities multiplied by world prices and divided by population. This treatment is especially important as prior empirical work relies on export data to measure resource extraction. However, exports do not capture the possible appropriated assets by the elites, hence understating the available resource rents. By using production data directly, we hope to relax this assumption. Moreover, to capture output from forestry extraction, we include Forestry Rents as a percentage of GDP. Political stability, rule of law, and trade openness enter as controls for the institutional and conditions emphasised in broader literature.

The specification includes interaction terms to test for non-linearities: *HCI* $\times$ *Production* and *GFCF* $\times$ *Production*, *HCI* $\times$ *Forestry Rents*, and *GFCF* $\times$ *Forestry Rents*. A positive coefficient on either would indicate that capital accumulation is associated with larger gains in economic complexity at higher levels of resource production, consistent with the hypothesis that the returns to converting resource wealth into productive capital are greatest where extraction is most intensive. Interaction terms are computed on mean-centred logged variables.

Our identification strategy relies on country-clustered standard errors. Standard OLS assumes independent and identically distributed errors; however, in panel data, observations from the same country across years are likely correlated. Ignoring this would produce artificially small standard errors and inflated t-statistics, leading to spurious significance. Ideally, we would also include country fixed effects to address omitted variable bias from time-invariant, country-specific characteristics. However, the limited number of country clusters in our sample makes it infeasible to implement both simultaneously. Faced with this trade-off, we prioritised accurate inference through clustering over correcting for potential coefficient bias through fixed effects. Therefore, results should be interpreted with the caveat that coefficient estimates may be biased by unobservable, time-invariant country characteristics.

Because ECI is highly path-dependent, a country's productive capabilities in one year are strongly correlated with its capabilities the following year. As a result, the regression includes a one-period lag of ECI. As a robustness check for reverse causality, we re-ran the regression using all lagged variables. The following equations describe Models (1) and (2):

$$
\begin{aligned}
ECI_{it} = \beta_0 + \beta_1 HCI_{it} + \beta_2 GFCF_{it} + \beta_3 Production_{it} + \beta_4 PoliticalStability_{it} \\
+\beta_5 RuleOfLaw_{it} + \beta_6 Trade_{it} + \beta_7 ForestryRents_{it} \\
+\beta_8 HCI_{it} \times Production_{it} + \beta_9 GFCF_{it} \times Production_{it} \\
+\beta_{10} HCI_{it} \times ForestryRents_{it} + \beta_{11} GFCF_{it} \times ForestryRents_{it} + \epsilon_{it}
\end{aligned}
\tag{3.1}
$$

$$
\begin{aligned}
ECI_{it} = \beta_0 + \beta_1 HCI_{it} + \beta_2 GFCF_{it} + \beta_3 Production_{it} + \beta_4 PoliticalStability_{it} \\
+\beta_5 RuleOfLaw_{it} + \beta_6 Trade_{it} + \beta_7 ForestryRents_{it} \\
+\beta_8 HCI_{it} \times Production_{it} + \beta_9 GFCF_{it} \times Production_{it} \\
+\beta_{10} HCI_{it} \times ForestryRents_{it} + \beta_{11} GFCF_{it} \times ForestryRents_{it} \\
+\beta_{12} ECI_{i,t-1} + \epsilon_{it}
\end{aligned}
\tag{3.2}
$$

3.4.3 Results

Table 3.3 reports the estimates across the three specifications. The Human Capital Index enters positively and significantly in every model, showing the effect is robust. The coefficient falls once lagged ECI is introduced (from 1.643 in Model 1 to 0.286 in Model 2 and 0.278 in Model 3) because much of the cross-country variation in human capital is absorbed by the lagged term, which captures the accumulated stock of productive capabilities. HCI's remaining statistically significant even after adding this variable indicates that human capital accumulation contributes to complexity growth beyond what is already embedded in a country's existing trajectory. Lagged HCI is significant, showing that previous levels of Human Capital increase ECI in the following period. Gross Fixed Capital Formation follows a similar pattern, entering positively and significantly across all specifications.

Production Value, our measure of resource extraction intensity, is negative throughout all models but none of the coefficients is significant. The direction is consistent with the literature: among resource-rich countries, higher per-capita extraction is associated with lower complexity, though the lack of significance suggests this direct effect is negligible once other variable effects are accounted for. Forestry Rents follows a similar pattern.

None of the four interaction terms is significant across the three specifications. However, three of the four interaction terms are positive across all specifications. A positive coefficient suggests that the association between human capital/gross fixed capital and economic complexity is stronger in countries with more intensive resource production, consistent with Venables (2016) argument that human capital investment yields larger returns precisely where the gap between existing capabilities and the requirement for diversification is widest. The results suggest that the composition of human capital and physical investment matters more at the aggregate level regardless of the resource dependence.

Among the controls, institutional variables (political stability, rule of law, trade openness) are not individually significant in Models (2) and (3), likely because within-country variation between 1995–2019 is insufficient to generate a detectable signal once accounting for ECI trajectory. The lagged dependent variable dominates the explained variance in Models (2) and (3), with R^2 rising from 0.371 in Model 1 to 0.787 in Models 2 and 3. The all-lagged specification, which lags all regressors by one period to address simultaneity, produces closely consistent results; HCI and GFCF remain significant at comparable levels, while natural resource production variables and all the interaction terms remain non-significant.

We also estimated a model with the change in ECI as the dependent variable as a robustness check. Results are reported in Appendix D. The results confirm the robustness of our main findings. Despite a notable decrease in R^2, from 0.789 to 0.095, the coefficients and standard errors of the variables of interest remain stable in sign, magnitude, and significance.

Economic Complexity in Resource-Rich Countries

	Dependent variable: Economic Complexity Index		
	(1)	**(2)**	**(3)**
Human Capital Index (log)	1.643***	0.286***	0.278***
	(0.406)	(0.106)	(0.107)
Gross Fixed Capital Formation (log)	0.147***	0.038***	0.034***
	(0.055)	(0.010)	(0.012)
Production Value (log)	-0.040	-0.008	-0.005
	(0.032)	(0.008)	(0.008)
Political Stability	0.028	0.013	0.013
	(0.069)	(0.019)	(0.017)
Rule of Law	0.346*	0.067	0.067
	(0.202)	(0.046)	(0.046)
Trade Openness (% of GDP)	0.001	0.000	0.000
	(0.003)	(0.001)	(0.001)
Forestry Rents (% of GDP)	-0.027*	-0.005	-0.003
	(0.016)	(0.004)	(0.003)
Interaction terms			
HCI × Production (log)	0.238*	0.035	0.039
	(0.141)	(0.029)	(0.029)
GFCF × Production (log)	0.020	0.007	0.004
	(0.030)	(0.006)	(0.006)
HCI × Forestry Rents	-0.080	-0.023	-0.018
	(0.072)	(0.017)	(0.018)
GFCF × Forestry Rents	0.013	0.006*	0.005
	(0.015)	(0.004)	(0.004)
Lagged dependent variable			
Economic Complexity Index (t−1)		0.811***	0.813***
		(0.044)	(0.044)
Constant	-2.938***	-0.558***	-0.565***
	(0.534)	(0.136)	(0.149)
Observations	1,324	1,270	1,270
Countries	54	54	54
R^2	0.376	0.789	0.789
Adjusted R^2	0.371	0.787	0.787
Clustered SE		Country	

Note: *p<0.1; **p<0.05; ***p<0.01. Clustered standard errors by country in parentheses. Model (1): contemporaneous regressors, no lagged dependent variable. Model (2): contemporaneous regressors with ECI(t−1). Model (3): all regressors lagged one period (t−1) to address endogeneity.

Table 3.3. Regression results

3.5 Machine Learning Approach

The econometric framework is developed by reviewing economic theory and research on the role natural resources play in economic development. The choice of predictors is therefore decided before the model is estimated. This guards against p-hacking (the practice of testing many specifications until a significant result appears) and ensures interpretable coefficients. On the other hand, variables which matter empirically but lack a clear theoretical motivation may be excluded. As a result, the model becomes susceptible to misspecification of functional form and omitted variable bias (Belloni et al., 2014).

We therefore complement the econometric analysis by using Machine Learning (ML) techniques. Our models of choice (LASSO, Ridge, Elastic Net, and Random Forest) take a more agnostic approach to variable selection. These models are able to test the predictive power of a broad set of candidate predictors, testing their relationship with economic complexity, as well as their reliability when applied to new data. To avoid reliance on outliers or spurious relationships, we utilise regularisation methods (LASSO, Ridge, Elastic Net) that shrink or eliminate coefficients associated with weak predictors, and ensemble methods like Random Forest, that average the predictions of many individual decision trees to reduce instability. Together, these tools serve as a check on the econometric findings. When the same variables appear important across both methods, the finding is less likely to be a result of model choice.

3.5.1 Regularisation Techniques

All regularisation techniques discussed here share the same linear modelling framework as Ordinary Least Squares (OLS). They assume a linear relationship between the dependent variable and the predictors, estimate coefficients by minimising deviation from the fitted curve and test how much this fitted line deviates from the actual data.

While OLS picks the coefficients that best fit the data, regularisation techniques are designed to decrease the size of coefficients whose relationship with the outcome variable is spurious. OLS might perform well on the original data, but it might not perform as well for new observations. LASSO, Ridge, and Elastic Net address this by adding a penalty term to each coefficient size, discouraging large coefficients and thereby reducing model complexity, allowing only variables with strong predictive power to influence the model.

The first of these models, LASSO, penalises the absolute value of the coefficients in the regression. Because of this, LASSO can shrink some coefficients exactly to zero, effectively performing variable selection and producing a sparse model. On the other hand, Ridge regression penalises the squared magnitude of the coefficients added in the regression. This shrinks all coefficients toward zero in a smooth and continuous way, reducing variance and improving stability. However, unlike LASSO, Ridge does not set coefficients exactly equal to zero, so all predictors remain in the model. Finally, Elastic Net uses a combination of the penalties of both LASSO and Ridge. This avoids LASSO's tendency to arbitrarily pick one variable from a correlated group and discard the rest, and Ridge's tendency to keep all of them with similar coefficients. Instead, Elastic Net tends to jointly shrink and select groups of correlated variables, producing more stable and interpretable coefficients (Zou & Hastie, 2005). LASSO may produce less stable selections among correlated predictors, Ridge may retain redundant information, and Elastic Net is expected to offer the best balance. We assess this empirically in the model comparison that follows. See Appendix E for further technical details.

Given that the general performance of regularised models is made more unstable by variables being extremely similar to each other, we test multicollinearity by analysing the Variance Inflation Factors (VIF) of each of them. The VIF measures how much the variance of a coefficient estimate is inflated due to correlation with other predictors; values above 10 are generally considered severe (Hair et al., 2019; Kutner et al., 2004). Our results (see Appendix E) confirm that no variable exceeds this threshold. This rules out the need to drop variables on collinearity grounds alone.

3.5.2 Robustness Check: Random Forest

The linear specifications common to regularisation models might constrain the analysis, so we also run the data through a Random Forest (RF) model. RF is an ensemble method that runs the same specification of the model on different subsamples of the full data, each time using a random subset of predictors, then averaging all of the runs to find the one that better predicts the data. Because it splits the predictors at precise values (i.e. ECI drastically higher when life expectancy > 70), it can capture non-linearities and interaction effects that a linear model would miss (Breiman, 2001). We set the number of trees to 500 and tune the number of candidate features per split (m) via cross-validation. Further details are available in Appendix E.

3.5.3 Predicting Best Performing Countries

Our final analysis focuses on finding the most predictive determinants of economic complexity, to then generate forward-looking projections for the economies in our sample.

First, models are trained using data from the 1995–2014 period and then they are evaluated under the last five years of the sample (2015–2019). This allows us to quantify out-of-sample predictive accuracy on two prediction targets: the ECI level and the year-on-year change in ECI ($\triangle$ECI).

Then, all four models are retrained on the complete 1995–2019 sample to maximise the information available for forward projection. Future values of each feature are extrapolated country-by-country using the linear trend estimated from the final five years of observed data. This approach assumes that recent structural trajectories persist at their current pace, which is a reasonable baseline assumption over a medium-term horizon, but might change as a result of exogenous shocks.

Forecasts are generated iteratively: for each country, the model predicts ECI for 2020 using 2019 features and the observed 2019 ECI as the lagged input; the 2020 prediction then feeds as the lagged ECI for the 2021 prediction, and so on through 2030. This process is carried out independently for both the level models and the $\triangle$ECI models (where predicted changes are accumulated from the 2019 baseline). The final ensemble prediction for each country-year is the simple average across all models, reducing the impact of any single specification's biases.

3.5.4 Results

Model agreement. Several variables consistently appear on the importance rankings (Figure 3.8). Domestic Credit to the private sector, a proxy for financial development, emerges as the strongest consensus predictor across all three estimators, with Human Capital and its interaction with Production fourth and second, respectively. This suggests that human capital's association with complexity strengthens with resource extraction intensity, similar to the results in our econometric specification.

Access to Electricity ranks third, though with wider dispersion across models consistent with Ridge's tendency to spread importance among correlated predictors. Urban population also seems somewhat important, possibly reflecting that agglomeration and population density are related to more complex products.

By contrast, variables such as Trade, Capital Formation, Landlocked, and the interaction of Rule of Law with Production receive near-zero importance from LASSO and Elastic Net, indicating they contribute little predictive power once the top-ranked features are accounted for.

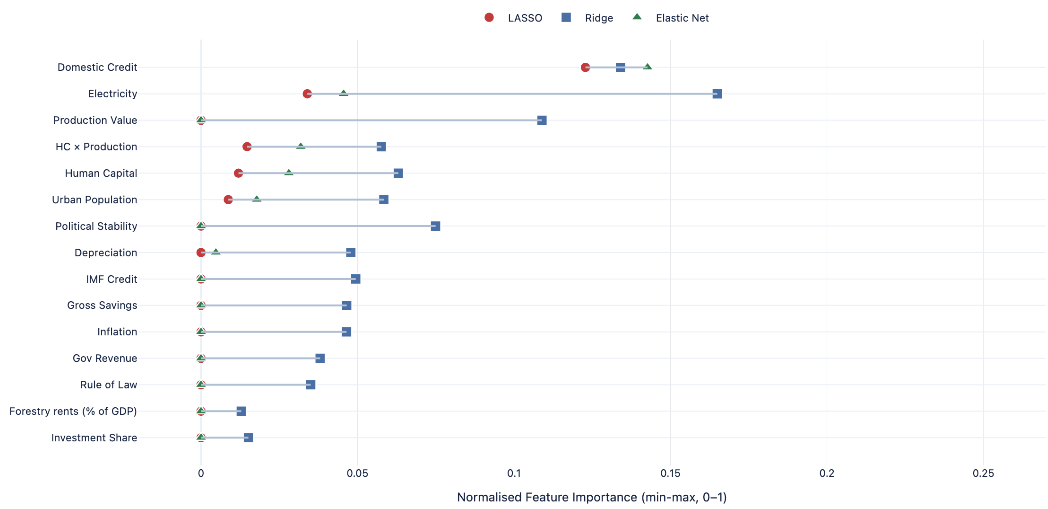

Figure 3.8. LASSO, Ridge and Elastic Net model agreement

Models' coefficients. All three models agree on the sign of every retained coefficient (Figure 3.9), suggesting the estimated directions of effect are not model-specific. Domestic Credit carries the largest positive coefficient, consistent with financial development facilitating productive diversification.

Human Capital, the Human Capital $\times$ Production interaction and Access to Electricity are also positively signed throughout, as is Urban Population, aligned with the expectation that agglomeration economies in urban areas support more complex production. On the negative side, Production Value is strongly negative, suggesting natural resource endowments might reduce Economic Complexity. Regarding variable selection, LASSO and Elastic Net retain 6 of 23 features, with the variables dropped by Elastic Net also carrying near-zero coefficients in Ridge, suggesting they hold little independent predictive power once the core determinants are accounted for.

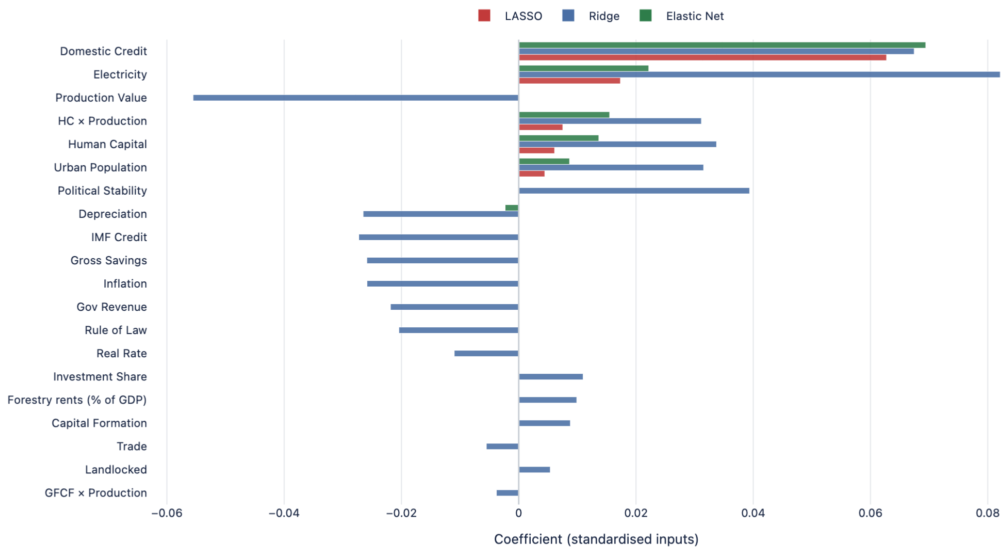

Figure 3.9. LASSO, Ridge and Elastic Net coefficients

Random Forest. Figure 3.10 presents the Random Forest feature importance rankings, based on Mean Decrease in Impurity. The Lagged ECI dominates by a considerable margin, as expected given the high persistence of economic complexity over time. Among the remaining variables, Domestic Credit, Human Capital and its interaction with Production and Access to Electricity, again emerge among the most important predictors, similarly to the regularisation models. Rule of Law, Production and the interaction between both also rank prominently as second-tier variables.

The convergence between the Random Forest and the linear estimators, despite different modelling assumptions, suggests the core results are not sensitive to model specification. Two additional robustness checks, a bootstrap analysis and a Leave-One-Country-Out (LOCO) procedure, are reported in Appendix F.

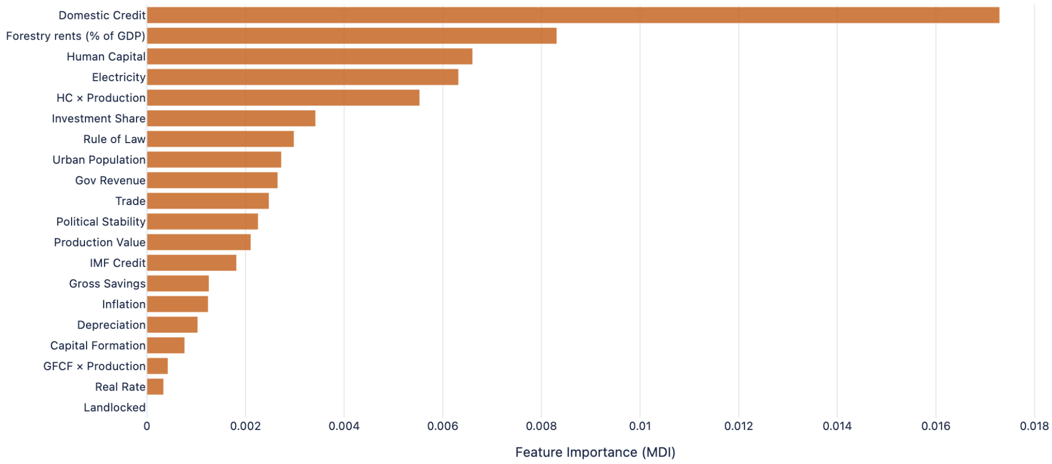

Figure 3.10. Random Forest feature importance

Prediction models and explained variance. Next, we check how reliable the predictive power of our models is. Figure 3.11 compares train and test R^2 across all four models, for both ECI levels and changes in ECI. For ECI levels, all models achieve test R^2 values between approximately 0.84 and 0.87, with the three regularised linear models performing comparably. Also, the narrow gap between train and test R^2 for the linear models indicates minimal overfitting.

On the other hand, for changes in ECI, explained variance drops substantially across all models, with test R^2 values falling below 0.15. This is expected as year-on-year changes in complexity are considerably noisier and harder to predict than levels, which benefit from the strong persistence captured by the lagged dependent variable.

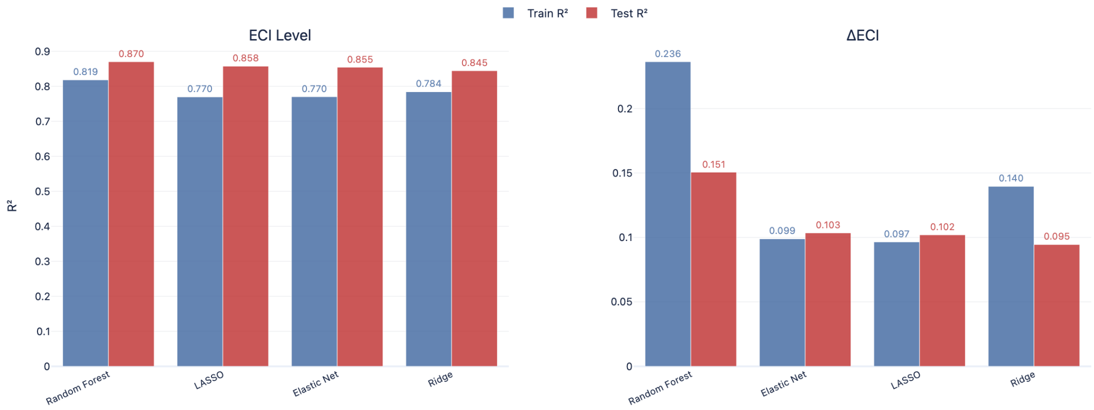

Figure 3.11. Train and test R^2 across models for ECI levels and changes in ECI (ΔECI)

Prediction accuracy. Figure 3.12 plots the Elastic Net predicted ECI against actual values for the test period (2015–2019) for both specifications. For ECI levels (left panel), the OLS fit line has a slope of 0.88, close to the ideal of 1, indicating that the model captures most of the cross-country variation in complexity levels. For changes in ECI (right panel), predicted values collapse toward zero regardless of the actual change observed, yielding a slope of just 0.16.

The largest variations from the model's baseline performance (residuals) come from countries whose complexity deviates from what their observable characteristics would suggest. Chad and Tanzania are consistently overpredicted, likely reflecting conflict, institutional disruptions, or large informal economies. Iraq and Mali's presence is likely dependent on previous or ongoing conflict and extractive-sector-dependent export bases. Our projections of economic performance should be read as indicative rankings of countries' potential to achieve greater complexity, rather than precise forecasts.

Predicting best performing countries. After retraining all our models using the complete 1995–2019 sample, we predicted ECI values for the 2020–2030 period. The final prediction for each country-year is the simple average across all four model-target combinations.

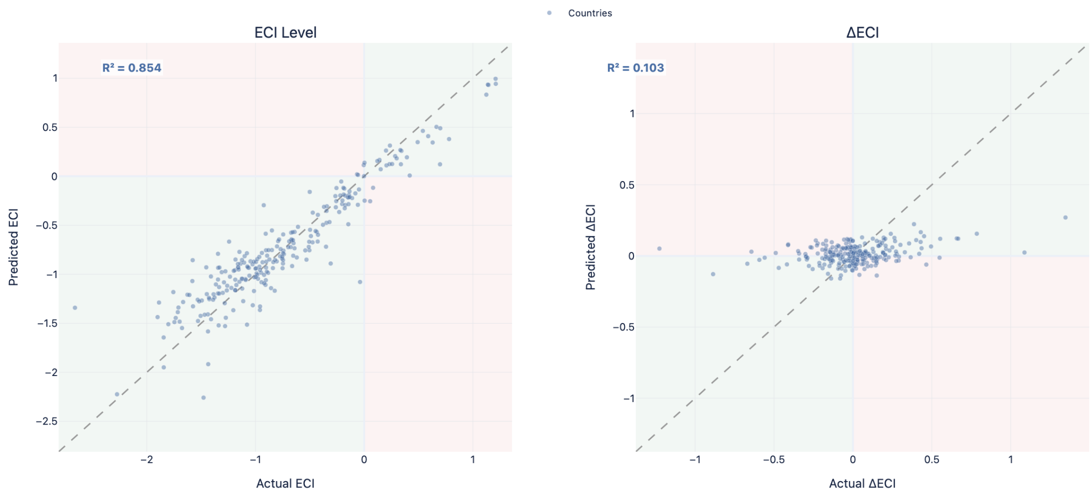

Figure 3.12. Elastic Net predicted vs actual ECI on the test set (2015–2019)

Figure 3.13 presents the projected ECI trajectories for countries with the largest predicted improvements and declines, respectively. Among the improving group, Ecuador (ECU), Equatorial Guinea (GNQ), and Mongolia (MNG) stand out as countries with a projected upward trend. These three countries are projected to converge toward higher complexity from lower starting points, a pattern consistent with their diversification efforts. Ecuador has used its strategic geographic position and expanding trade agreements to move beyond oil, relying instead on expanding renewable energy infrastructure (World Bank, 2023). Mongolia has invested heavily in education and financial infrastructure while pursuing mining-linked industrialisation (IMF, 2023a). Equatorial Guinea, starting from a very low complexity base, is predicted to diversify, possibly as a result of ongoing improvements in its governance and education.

Among the declining group, Zimbabwe (ZWE), Saudi Arabia (SAU), and Libya (LBY) are projected to experience the biggest decrease in ECI, showing limited diversification capacity. Saudi Arabia's projected decline is particularly notable given that it invested billions of dollars into the ambitious Vision 2030 reform programme (IMF, 2024a). While this might be a sign that oil-dominant economies struggle to diversify despite favourable policies, it might also be the product of the model not being able to observe changes in policy. On the other hand, prolonged political fragmentation and recurring armed conflict have left Libya's economy almost entirely dependent on crude oil exports, with hydrocarbons accounting for over 95% of export revenues and approximately 60% of GDP (World Bank, 2023d). Finally, Zimbabwe's trajectory is shaped by governance and macroeconomic instability that hamper the preconditions needed for complexity gains (IMF, 2025a).

Figure 3.13. Projected ECI trajectories, top improving and declining countries (2020–2030)
Note: Solid lines show observed ECI (1995–2019); dashed lines show model projections.

3.6 Conclusions

The ML analysis identifies three main variables associated with a higher complexity. These are financial development (proxied by Domestic Credit), infrastructure capacity (proxied by Electricity Access), and human capital, with its interaction with the level of natural resource production. These results complement the econometric findings in two ways. First, it unveils important predictors that were not considered in the econometric framework. Domestic Credit, for instance, emerges as the strongest predictor despite its absence from the regressions. Meanwhile, variables that are insignificant in the econometrics section, such as trade openness and political stability, receive near-zero ML importance as well, confirming their low predictive value rather than suggesting their effects were simply absorbed by fixed effects.

Finally, one notable divergence concerns physical capital, where gross fixed capital formation is significant econometrically but ranks low in the ML results. This likely reflects the fact that domestic credit and electricity access already capture broad-based infrastructure investment, whereas GFCF in hydrocarbon economies is dominated by resource-extractive capital deepening.

Chapter 4

Case Analysis

These three structural pillars identified in the quantitative analysis motivate the selection of three case studies: the Republic of the Congo, Azerbaijan, and Chile. The Republic of the Congo acts as a representative case for those oil-exporters that jointly lack financial depth, infrastructure, and human capital. Azerbaijan serves as a case wherein resource-led economic growth is achieved at the expense of crowding out non-oil tradables through Dutch Disease dynamics, accentuated by poor institutional quality. The case of Chile showcases the importance of strong institutions and infrastructure. Yet these factors alone do not solve its declining complexity trajectory, where human capital and innovation capacity remain as bottlenecks. Together, these cases examine which constraints bind at different levels of development, presenting qualitative evidence shaping value-added industries. Table 4.1 summarises key indicators for the three countries.

Case Studies Comparison: 1995 vs 2019

	Congo, Rep.		Azerbaijan		Chile		Sample Average	
	1995	**2019**	**1995**	**2019**	**1995**	**2019**	**1995**	**2019**
Continuous variables								
Population	2,749,461	5,616,661	7,684,850	10,024,283	14,527,381	19,197,744	23,893,828	38,262,461
GDP Per Capita	4,912.43	4,093.99	3,142.33	14,541	12,941	23,880	10,970	14,076
Inflation (annual %)	9.42	2.21	411.76	2.61	8.23	2.56	97.37	14.12
Economic Complexity Index (ECI)	-0.83	-1.08	-0.15	-0.83	-0.08	-0.41	-0.72	-0.76
Human Capital Index	1.90	2.16	2.92	3.35	2.71	3.15	1.85	2.33
Gross Fixed Capital Formation (% of GDP)	21.96	18.86	3.68	12.42	18.51	23.48	13.58	18.00
Political Stability	-1.23	-0.30	-0.83	-0.69	0.68	-0.02	-0.54	-0.69
Rule of Law Index	0.47	0.15	0.05	0.04	0.96	0.94	0.35	0.37
Natural Resource Production Value Per Capita	472.50	618.86	234.15	952.33	620.56	1,273.47	929.27	992.70
Trade (% of GDP)	128.31	91.58	85.89	85.82	54.69	57.61	65.36	70.30
Access to Electricity (% of pop.)	29.40	47.70	97.00	100.00	95.76	100.00	52.98	69.95
Gross Savings (% of GNI)	-3.97	38.98	7.29	29.72	25.19	20.57	21.41	25.01
Domestic Credit to Private Sector (% of GDP)	8.11	12.64	32.87	23.00	73.08	124.88	19.54	32.96
Binary variables (share = mean)								
Hydrocarbons Dominant (=1)	100.0%	100.0%	100.0%	100.0%	0.0%	0.0%	53.7%	51.9%
Subsoil Metals Dominant (=1)	0.0%	0.0%	0.0%	0.0%	100.0%	100.0%	16.7%	16.7%
Precious Metals Dominant (=1)	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	25.9%	25.9%

Note: Sample Average represents the mean of high resource countries for the respective year.Binary variables report the share of country-year observations equal to 1.

Table 4.1. Case studies comparisons: 1995 vs 2019

4.1 Republic of the Congo

4.1.1 Overview

The Republic of the Congo (ROC) is a lower-middle-income country located in Central Africa, with access to the Atlantic Ocean. It has a population of 6.3 million people, 64% of whom live in urban areas (World Bank, 2023). Its biggest city is Brazzaville, the capital, located directly across from the capital of the Democratic Republic of Congo (Kinshasa). Pointe-Noire, the second biggest urban centre, hosted the country's first oil extraction site and currently functions as its main key logistics and commercial hub.

With proven reserves of approximately 1.8 billion barrels and a current output of around 270,000 barrels per day, oil has dominated the Congolese economy since the early 2000s. In 2025, it accounted for roughly 40% of GDP, 80% of total exports, and 60% of government revenues (World Bank, 2023a; 2025a). The country also holds substantial reserves of natural gas, estimated at 10 trillion cubic feet (EIA, 2024), as well as other mineral resources that remain largely unextracted. This includes an estimated 25 billion tons of iron ore and 3.2 billion tons of potash (World Bank, 2025).

4.1.2 Why We Chose This Country

The Republic of the Congo is a typical example of a resource-rich economy that has struggled to utilise its natural wealth to diversify and improve its industrial capacity. The country combines high hydrocarbon endowments, vast amounts of forestry, and largely unutilised mineral reserves with severe underdevelopment and extractive institutions.

Its economy remains extremely volatile, mainly due to its reliance on oil: as the country approached upper-middle-income status during the oil boom in 2014, the price of the barrel dropped significantly. This generated a severe recession from 2015 to 2021, with government revenues falling from 37.5% of GDP in 2014 to 12.3% in 2020 (World Bank, 2023a; IMF, 2023b). As a result, public spending on education and health strongly depends on foreign economic cycles, so it is usually pro-cyclical, restricting long-term human capital accumulation.

The inability to successfully accumulate wealth and provide education to its citizenry, together with macroeconomic volatility, makes the ROC a case study to test the results of our model. When inadequate institutions cannot sustain the instability generated by shifting commodity prices, diversification cannot take place, the state is not able to provide basic services and carry out investment all at the same time.

4.1.3 Economic Structure and Trajectory

As previously established, the country's economy is highly reliant on the oil industry. In 2024, oil and its derivatives accounted for 67% of total exports, mainly through crude rather than refined petroleum (63.8% compared to 3.8%). While copper is the second export by size (at 22%), this is mostly a result of re-exports of refined copper transiting from the DRC. This configuration makes it vulnerable to external shocks, both from oil prices, from its own neighbours' production and from its trade partner, as 49.3% of exports are towards China (Harvard Growth Lab, 2025a).

The construction and finance sectors tend to expand during oil booms but contract during downturns, reflecting pro-cyclical investment (IMF, 2024b). Given this context, it is very difficult for natural capital extraction to consistently generate fiscal revenues every year, so the ROC fails to develop the know-how necessary to extend supply chains.

This is reflected by ROC's low ECI ranking (124th out of 130 countries), the result of an export basket concentrated in low-technology, resource-based products with limited spillovers to other sectors. This reliance on oil has an additional side effect, as the hydrocarbon industry is capital-intensive and employs only a small share of the labour force. This leaves approximately 75% of workers in low-productivity informal employment (World Bank, 2023b).

Although the 2014 oil price collapse triggered a recession that lasted until 2021, the economy has since begun recovering, growing modestly in 2024 (2.6%) and is expected to expand further in the coming years (World Bank, 2025; IMF, 2024). Disruptions in electricity and fuel supply have, however, increased domestic production costs, further depressing industrial development. Inflation is projected at 4.0% in 2025, above the CEMAC regional target of 3.0%, driven in part by energy supply constraints and rising food prices (World Bank, 2025b).

4.1.4 Human Capital

Investment in education has historically been highly volatile (World Bank, 2025). While the HCI has slightly increased from 1.9 in 1995 to 2.16 in 2019, it has become lower compared to other resource-rich economies. This is likely to be the result once again of procyclical educational spending, oscillating between 8.4% and 1.6% of GDP between 1978 and 2005. Teacher shortage remains an issue. The ROC currently has roughly 23,000 teachers against an estimated need of 48,000, and only 60% of children who complete primary school transition to secondary education (Broken Chalk, 2020).[1]

Most of the country's labour force is employed in low-productivity non-oil sectors, while a small portion of the population works in the highly productive hydrocarbon sector, as a result of its low labour intensity. In 2020, 98% of the 75,118 active firms were micro or small enterprises, and about 94% were informal (World Bank, 2023). This informal-

[1] The Penn World Table Human Capital Index measures how education translates into worker productivity, where higher values indicate more productive workers due to education.

ity restricts its value-added activities, as informal units are excluded from formal credit. Consistent with this, in 2024 only 8.5% of firms used banks to finance their investments according to World Bank data. As a result, in-firm technological adoption, workforce upskilling and the productivity gains struggle to take place, further straining economic diversification.

4.1.5 Infrastructure and Technology

The country's underdeveloped infrastructure system represents a binding constraint on its ability to diversify. Frequent outages from aging transmission networks increase production costs and discourage investment in manufacturing and agro-processing (World Bank, 2023b), while only half of the population has access to electricity. There is also a stark regional divide, with access to electricity reaching 67% of people in urban areas, while only 12% of those in rural areas (World Bank, 2025).

The road networks are sparse, as only 13% of roads are paved (CIA, 2024), while the state-owned CFCO rail system operates well below capacity and weak domestic production linkages leave Pointe-Noire's port underutilised (World Bank, 2023). Digital infrastructure also remains underdeveloped despite the government's ambition to expand the digital economy. While mobile connections are near-universal (99.8% of the population), only 38% of the population uses the internet, reflecting limitations in connectivity and digital inclusion (DataReportal, 2026).

Finally, the financial infrastructure system remains small relative to the size of the economy, and is highly concentrated in commercial banks, which account for approximately 84% of total financial sector assets (IMF, 2025). Access to finance is largely concentrated in urban areas, with only 18% of adults holding a formal account as of 2021 (IMF, 2025).

4.1.6 Governance and Institutions

Beyond these economic and infrastructure constraints, the institutional environment challenges the diversification process. The ROC is a presidential republic with highly concentrated executive power and authoritarian governance, led by President Denis Sassou Nguesso. He has ruled the country since 1979 (except from 1992 to 1997) and has won all elections since returning to power. The country's governance indicators reflect this: V-Dem classifies it as an electoral autocracy, Freedom House rates it 'Not Free' with a score of 18 out of 100, and it ranks 123rd out of 143 countries on the Rule of Law Index (Freedom House, 2025; V-Dem Institute, 2024; WJP, 2024).

Oil revenues are highly centralised and have historically lacked full transparency in their allocation and management. Although governance reforms have been introduced under IMF-supported programs since 2022, public financial management challenges persist. Oversight of stakeholders in the oil sector supply chain is still not effectively monitored, as the national oil company (SNPC) reports import invoices that are neither verified nor challenged by the regulatory agency (ARAP) or the Directorate General of Hydrocarbons (IMF, 2024a).

4.1.7 Peer Comparisons

Despite significant oil wealth, ROC underperforms relative to other countries with similar resource endowments (Figure 4.1). This is primarily driven by high fiscal revenue volatility, proper for an oil-dependent country with low institutional quality.

Angola: same export structure, but with different outcomes. Both countries share similar economic characteristics, as they rely heavily on hydrocarbons. Angola is the second-largest oil producer in Sub-Saharan Africa, while the ROC ranks third. In the early 1990s, both countries had comparable GDP per capita levels. Since then, Angola has reached significantly higher income levels, while the ROC's real GDP per capita has declined to 1970 levels (World Bank, 2025). The divergence is not simply a matter of export composition. Angola produces approximately five times more oil than the ROC, which provided a much larger fiscal revenue base and allowed the government to finance a massive post-war infrastructure reconstruction programme, channelling billions into roads, energy, and transport networks, much of it backed by oil-collateralised credit lines from China (Alves, 2013).

Azerbaijan: physical capital accumulation from oil extraction. Since 1995, Azerbaijan's stock of produced capital has expanded by 428%, while the ROC's has increased by only 40% (World Bank, 2023). Infrastructure indicators reflect this gap: obtaining an electricity connection in Azerbaijan takes 41 days and costs 126% of income per capita, whereas in the ROC it takes 134 days and costs 5,569% of income per capita. Azerbaijan scores 7 out of 8 on reliability of supply and tariff transparency, while the ROC scores 0 (World Bank, 2023). These differences show differential impacts on firm-level competitiveness and the feasibility of non-oil industrial development.

Malaysia: diversification through human capital development investment. Among aspirational peers, Malaysia illustrates the role of strategic human capital investment. Its national oil company, Petronas, has supported technical education, vocational training, and research initiatives, generating broader economic spillovers. In contrast, the ROC's Société Nationale des Pétroles du Congo has functioned primarily as a revenue collection mechanism, with limited reinvestment in skills development (World Bank, 2023).

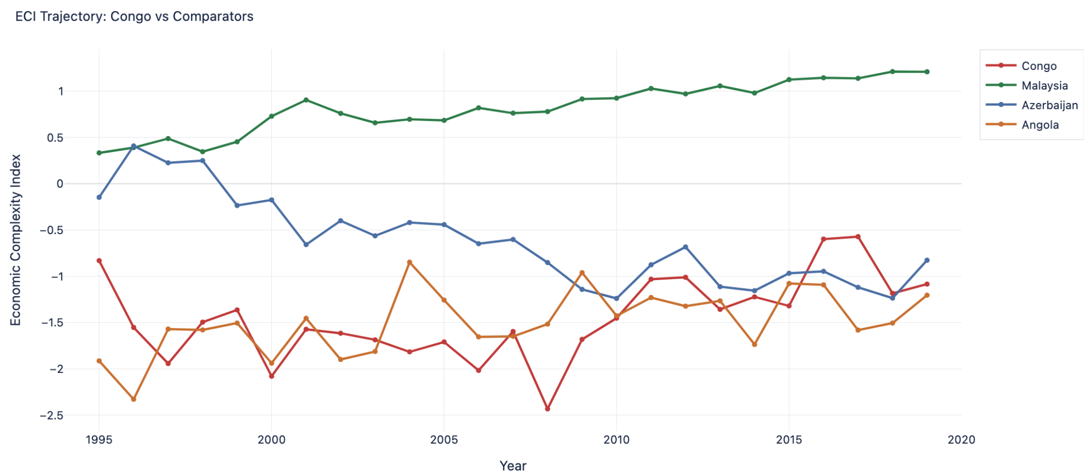

Figure 4.1. ECI trajectory: Republic of the Congo and comparators

4.1.8 Policy Recommendations

ROC has experienced pro-cyclical fiscal patterns, debt distress, and weak wealth accumulation. Although the National Development Plan (2022–2026) provides a strategic path toward diversification, implementation capacity remains limited.

Strengthening human capital accumulation. Currently, Marien Ngouabi University is the sole state-funded tertiary institution in the country, showcasing large supply-side constraints to education. Improving human capital outcomes requires stabilising and increasing investment in education and health, which has historically been volatile and closely tied to oil revenues. On a broader note, there is a fundamental coordination problem as teacher shortages in secondary schooling are linked to low student admission and attendance rates (World Bank, 2020c). Government policies should focus on continuous student progression and improving educational quality rather than broad-based, low-capacity schooling. Moreover, ROC's low labour productivity highlights the need for vocational education in sectors such as agriculture, logistics, and construction, allowing for better allocation of labour, and decreased attractiveness of low-productivity informal activities among youth.

Fiscal discipline and resource revenue management. The ROC's fiscal position is highly procyclical, further amplifying macroeconomic volatility. Adopting a credible fiscal rule anchored to a structural oil price benchmark would help delink expenditure from short-term commodity fluctuations. This should be complemented by the establishment or strengthening of a sovereign wealth fund mechanism within the Economic and Monetary Community of Central Africa (CEMAC) framework, allowing countercyclical spending and intergenerational savings.

Infrastructure systems. The electricity sector is a binding constraint on non-oil growth. Despite vast hydroelectric potential, the state-owned Énergie Électrique du Congo (E2C) reports distribution losses of over 50%, a national electrification rate of just 42% that falls to 5% in rural areas, and daily blackouts across Brazzaville and Pointe-Noire (USAID, 2017). Enabling private participation in distribution can alleviate the inefficiencies of the state monopoly. The 2003 Electricity Code provides the institutional framework necessary. The country must have the political will to enforce it to allow for greater access to electricity.

4.2 Azerbaijan

4.2.1 Overview

Azerbaijan is the largest and most populous country in the South Caucasus, classified as upper-middle-income, with around 10 million inhabitants. Economic activity and population are heavily concentrated around Baku (World Bank, 2024a). The oil and gas sectors dominate the country's economy, accounting for roughly 90% of export revenues and 50% of GDP (IEA, 2023; BP, 2024), making it one of the ten most fossil-fuel-dependent economies globally. Governance indicators are weak, with Azerbaijan scoring in the bottom quartile on V-Dem's political corruption index, consistent with a consolidated authoritarian regime.

Fiscal management is anchored by the State Oil Fund of Azerbaijan (SOFAZ), a sovereign wealth fund established in 1999, though transfers to the state budget have at times exceeded the fund's own savings rules, raising long-run sustainability concerns (IMF, 2023b). The country holds a Ba1/BB+/BB+ sovereign credit rating, reflecting both the stability generated by oil income, as well as its excessive reliance on hydrocarbons. Azerbaijan's post-independence trajectory follows the Asian Development Bank's (ADB) four-stage framework (Hampel-Milagrosa et al., 2020): post-Soviet contraction (1991–95), reform era (1995–2003), oil boom (2003–14), and shock and partial recovery (2014–19), with GDP growth slowing substantially since the 2014 oil price decline.

4.2.2 Why We Chose This Country

Azerbaijan's economy remains heavily dependent on oil and gas. Consistent with this, the country's ECI declined between 1995 and 2019, even as GDP growth averaged 6.59% over the same period. Its exporter destinations are concentrated, as Azerbaijan directs 44% of its oil exports to Italy alone, with Germany a distant second at 7%. Non-oil exports include agricultural products, aluminium, steel, and electricity. However, a significant share of diversified manufacturing (over 45%) consists of petroleum refining, derivatives, chemicals, rubber, and plastics. Almost half of non-oil exports are themselves related to hydrocarbons, suggesting the country is even less diversified than aggregate statistics suggest.

Despite continued reliance on hydrocarbons, geopolitical developments have opened new markets for Azerbaijani gas. Following Russia's invasion of Ukraine in February 2022, European countries moved rapidly to reduce dependence on Russian pipeline gas, positioning Azerbaijan as a strategically attractive alternative supplier. Azerbaijan has increased its gas exports to Europe by 56% since 2021. Azerbaijan's total gas exports reached 25.2 billion cubic meters in 2025, with more than half of the supply going to the European Union (Birbayeva, 2026). However, European gas prices have been volatile in recent years, caused by the war juncture, threatening the potential long-term sustainability (EBRD, 2024).

This tension between robust GDP growth and declining economic complexity contrasts with the cross-country findings presented in the preceding sections, positioning Azerbaijan as a compelling case for examining the extent to which resource-driven growth is sustainable in the long run.

4.2.3 Economic Structure

Azerbaijan's economy is primarily based on oil, with the rest of the sectors trailing behind. In 2024, non-oil sectors include trade and vehicle repair (10.7% of GDP), transport and warehousing (7.0%), construction (6.7%), agriculture (5.7%), tourism (2.4%), and information and communications (1.9%). Whilst Azerbaijan's economy includes these sectors, the ECI has decreased 0.68 points from 1995 to 2019. Although it advanced its ECI from 2018 to 2023, the measure remains relatively low in absolute terms, ranking the country 95th globally.

Fiscal and financial position. Azerbaijan's financial system has improved since the 2015 crisis. During the 2000–2014 period, the number of loans grew by more than 48 times, with the increase accelerating following the opening of the Baku–Tbilisi–Ceyhan (BTC) pipeline. Credit allocation stagnated during the 2008–2011 Global Financial Crisis but recovered over the following three years (CESD, 2016). However, the 2015–16 banking crisis led to widespread devaluations and a new recession, hurting bank profitability and exposing a host of bad debts as a result. Since then, authorities have begun closing non-viable banks and addressing the high incidence of Non-Performing Loans (NPLs) (IMF, 2016).

The rehabilitation of the state-owned International Bank of Azerbaijan (IBA) following the 2015–2016 crisis was largely successful. Through asset transfers totalling 37% of GDP and foreign currency liability restructuring, the IBA's financial health improved significantly. Though it remains dominant, it currently holds 26% of sector assets compared to 38% in 2015. The aggregate NPL ratio subsequently fell from 21% at end-2016 to 2.7% by mid-2024, indicating a decline in the prevalence of distressed lending.

Azerbaijan's pre-2014 growth was driven by rising energy exports, but declining oil prices caused GDP contractions of 3.1% in 2016 and 0.8% in 2017 (Moody's, 2025). Fiscal management has been highly procyclical, with SOFAZ spending at times exceeding the fund's own investment returns (Hampel-Milagrosa et al., 2020). Due to the country operating without any fiscal rule until 2018, surpluses turned into deficits, and the debt-to-GDP ratio tripled from 11% to 38% during the 2015–16 shock. The latter was mainly in the form of foreign-currency debt (IMF, 2016), making it less sustainable in the long run.

The fiscal rule, introduced in 2018, set upper limits on oil revenue spending based on net financial assets (Azertac, 2018), though governance weaknesses persisted. The government continued carrying out procyclical fiscal policies, and fiscal risks from state-owned enterprises persisted. The absence of binding authority within SOFAZ's Supervisory Board to reject presidential withdrawal orders also threatened fiscal sustainability (IMF, 2025).

4.2.4 Human Capital

Azerbaijan's Human Capital Index improved by 15% between 1995 and 2019 (from 2.91 to 3.35), though it remains below regional peers (World Bank, 2022a). The country is locked in a low-productivity, low-skill cycle. Weak non-oil labour demand suppresses returns to education. Domestic firms report inadequate skills and low digital literacy as key constraints to transformation, including a lack of experience with modern methods of production. Limited labour demand in productive sectors and skill gaps affect productive employment opportunities (World Bank, 2022a).

Relative to Azerbaijan's income level, children and youth suffer from relatively poor health and weak learning achievements, setting them up on a lower productivity path (World Bank, 2022a). High incidence of non-communicable diseases (NCDs), such as cardiovascular diseases and diabetes, also limits the older population's contributions to productivity and growth (World Bank, 2022a).

4.2.5 Selective Infrastructure Spending and Dutch Disease

Azerbaijan's infrastructural development since independence in 1991 has been focused on energy export infrastructure (pipelines). However, other infrastructure sectors have declined, and distortions were introduced by oil-financed spending, particularly Dutch Disease effects on non-oil sectors. Infrastructure spending has been mainly concentrated on export pipelines and in the capital Baku. The railway sector shows a contrasting story. Rail goods transported dropped from 26.5 million tons in 2006, at the peak of the economic cycle, to 22.4 million tons in 2010 and 15.5 million tons in 2016 (Asian Development Bank, 2021). The ADB identified structural institutional failures as the root cause. Decades of deteriorating infrastructure and rolling stock, as well as a lack of commercial orientation in management and planning decisions.

Also, the spending effect crowded out non-oil tradable sectors. Azerbaijan suffers from "relative de-industrialisation" in the non-oil tradable sector, while the non-tradable sector substantially expanded during the 2000–2007 period. Rapid increases in wages and non-tradable prices led to an appreciation of the real exchange rate (Hasanov, 2013). Also, a booming resource sector tends to draw capital and labour away from a country's manufacturing and agricultural sectors, leading to a decline in exports of agricultural and manufactured goods and inflating the cost of non-tradable goods (Hasanov, 2013).

4.2.6 Governance and Institutions

Azerbaijan's political stability index improved marginally between 1995 and 2019, while rule of law remained virtually unchanged. The political system is characterised by autocratic presidentialism with neo-patrimonial, political ties driving the state management, structures and extensive rent capture, common to many post-Soviet countries (Franke et al., 2009). The judiciary also lacks autonomy, as judges often collaborate with prosecutors and security agencies to target political opponents. The bar association is controlled by the executive, where lawyers representing members of the opposition face disbarment (Freedom House, 2024). The media is predominantly state-controlled, and independent journalists face criminal charges and imprisonment for investigating corruption.

This weak institutional framework has enabled rent-seeking, with the ruling party utilising state-owned fossil fuel enterprises to consolidate power through politically connected contracts. Opaque procurement processes allow elites to capture funds intended for climate resilience and renewable energy, discouraging diversification in sustainable energy (Transparency International, 2025).

4.2.7 Lack of Transparency and Income Inequality

Despite economic growth between 1995 and 2019, it remains unclear whether its benefits have been broadly shared. Azerbaijan's State Statistical Committee does not publish a Gini coefficient, and the World Bank's data covers only 1995–2005, ending precisely as oil revenues accelerated.

The most reliable post-2005 proxy, the World Bank's shared prosperity premium, has been negative since 2015: consumption among the bottom 40% fell nearly twice as fast as among the top 60% (World Bank, 2022b). This statistical opacity is itself consistent with the rentier-state governance framework established by Guliyev (2013) and Franke et al. (2009). Inequality is likely substantially higher than official figures indicate.

4.2.8 Peer Comparison

We compare Azerbaijan's trajectory with that of Uzbekistan, Malaysia, and Iraq (Figure 4.2). Malaysia, however, serves as a potential counterfactual. Both countries were upper-middle-income hydrocarbon economies in the mid-1990s with comparable ECI levels, yet Malaysia sustained a continuous upward trajectory to $+1.1$ by 2019 while Azerbaijan collapsed to -1.2 by the mid-2000s, a divergence mapping directly onto policy choices, as Malaysia channelled Petronas revenues into downstream manufacturing and electronics while Azerbaijan allowed petrodollars to crowd out non-oil tradables entirely.

Azerbaijan's ECI collapse is temporally linked to its oil boom, a clear Dutch Disease signal, with the sharpest deterioration occurring between 2000 and 2008, precisely as GDP grew 20% annually. The partial recovery to -0.88 by 2019, driven by the post-2015 oil price crash forcing some non-oil reorientation, remains modest: still roughly two full ECI points below Malaysia. Two binding constraints explain this underperformance relative to Azerbaijan's human capital and infrastructure levels: severely inadequate rule of law (consistently around -0.7 versus Malaysia's $+0.5$), which undermines the contract security needed for sophisticated investment, and extreme hydrocarbon concentration that crowds out non-oil tradable activity through Dutch Disease channels.

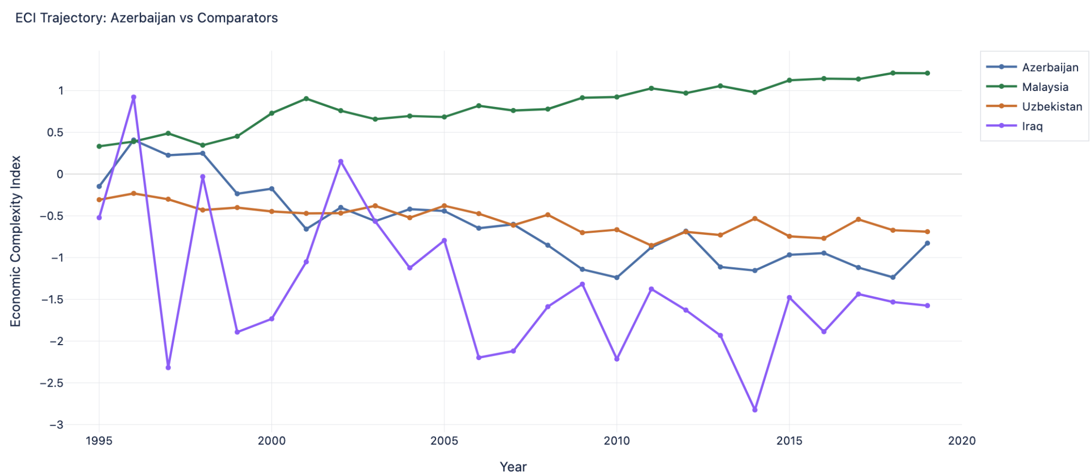

Figure 4.2. ECI trajectory: Azerbaijan and comparators

4.2.9 Policy Recommendations

Although Azerbaijan's oil wealth has enriched the state, its impact on living standards remains limited, with poverty, corruption, and repression persisting. High-skill sectors such as oil and gas, ICT, and advanced manufacturing are either dominated by foreign expertise, while the sectors employing the most workers (agriculture and trade) remain low-productivity (Emerging Europe, 2024).

Establish independent mechanisms to mitigate rent capture. Rent capture by ruling government political links directly suppresses private sector innovation and increases barriers to entry, disincentivising productive diversification. This calls for the establishment of oversight bodies for sovereign wealth fund governance (SOCAR and SOFAZ), transparency for large enterprises, competitive procurement in public contracting, and sufficient judicial independence to allow private firms to compete with state-linked enterprises.

Invest in human capital and productive diversification. Following Hidalgo and Hausmann (2009) framework, Azerbaijan should prioritise sectors closely related to its existing export basket: petrochemicals, agro-processing, logistics, and renewable energy engineering adjacent to oilfield capabilities. Direct investment through concessional financing and industrial park incentives can promote diversification and stop further oil-sector deepening. The World Bank's Azerbaijan Human Capital Review (2022a) identifies a self-reinforcing trap in which weak non-oil labour demand suppresses investment in education, which in turn perpetuates the demand gap. Breaking this cycle requires simultaneous action on skills supply and industrial policy. Without such intervention, potential growth remains constrained at roughly 2–2.5% (IMF, 2026), limiting the non-oil revenue base and the long-run sustainability of SOFAZ transfers.

Create independent institutions for economic data production and transparency. Understanding economic phenomena relies on being able to assess current states and trace the consequences of events (Einav and Levin, 2013). Accurate, granular data being available to private actors and researchers is a condition that Azerbaijan currently lacks. This requires an independent national statistics office with statutory protections and international methodological standards, alongside a fiscal council mandated to publish non-oil deficit projections and SOFAZ sustainability assessments.

4.3 Chile

4.3.1 Overview

Chile sits along the western coast of South America, with a population of approximately 19.5 million. It is the fifth-largest economy in Latin America by nominal GDP and is classified as high-income by the World Bank. Chile's mining wealth is heavily concentrated in the northern regions, while the south depends on lower-value forestry, agriculture, and salmon farming. This leaves most southern municipalities as net fiscal recipients, collectively gathering less than 13% of total tax income (OECD, 2017a).

The country is the world's largest copper producer and second-largest lithium producer, with mining contributing roughly 12% of GDP and 57% of exports (ITA, 2025; USGS, 2024). This concentration of natural resource wealth is paired with comparatively strong institutional quality. Since returning to democratic governance in 1990, Chile has maintained market-oriented policies and open trade. It also scores among the highest in Latin America on governance indicators.

Fiscal governance is similarly strong. A structural balance rule ties fiscal spending to long-run copper prices (Marcel, 2011), supported by two sovereign wealth funds (ESSF and PRF) and an independent inflation-targeting central bank, contributing to the highest sovereign credit rating in the region (A2/A/A-).

4.3.2 Why We Chose This Country

Chile satisfies some of the institutional prerequisites that the literature identified as potential binding constraints on diversification: democratic governance, solid fiscal position, and developed infrastructure. Despite this, its ECI has been declining since the mid-2000s, worker productivity sits 50% below the OECD average, and income inequality remains among the highest in the OECD (OECD, 2025; IMF, 2025). Whereas weak governance can plausibly explain low diversification in Azerbaijan or the Republic of Congo, this is unlikely to be the case for Chile. This dichotomy illustrates both the possibilities and the limits of sound institutions when broader diversification remains absent. Chile's critical mineral assets allow it to supply multiple battery manufacturing stages, from lithium refining to copper foil for anodes and wiring for EV motors. Rising global demand creates an opening to move beyond extraction, but doing so requires accumulating sufficient human capital and technical capability.

4.3.3 Economic Structure and Trajectory

Chile's commodity dependence stretches back to the nitrate boom following the War of the Pacific (1879–1883), which financed development but left the economy dangerously exposed, evidenced with the 87% fall in exports during the Great Depression (Thorp, 1984). Copper gradually replaced nitrates and became central to state revenues after CODELCO's creation in 1971. Post-1973 liberalisation and the democratic transition then fuelled average growth of 6.5% through the 1990s, with per capita income nearly tripling and poverty halving (Fuentes and Mies, 2005; World Bank, 2004).

Chile built a layered fiscal framework to manage commodity revenues countercyclically. Part of this stability rested on the Copper Price Compensation Fund (1985), which deposited surplus copper revenues when prices exceeded a long-term reference and released them when prices fell below, establishing the principle that commodity income should be spent counter-cyclically. The structural balance rule (2001) formalised this into a fiscal target, tying government expenditure to revenue estimated at potential GDP and long-term copper prices (Marcel, 2011). The 2006 Fiscal Responsibility Law then created the ESSF and PRF as dedicated savings vehicles for surplus mineral revenues.

This accumulation enabled Chile to inject USD 4 billion into the economy during the 2008 financial crisis, drawn from the USD 20 billion held by the ESSF at the time (Kumhof and Laxton, 2009).

Fiscal and financial position. Central government revenue has averaged 22–23% of GDP over 2019–2024, with copper-related fiscal income accounting for 3 to 5% of the total (IMF, 2025). The country holds one of the lowest public debt burdens in the OECD. Once sovereign fund assets are included, net government debt falls to approximately 30% of GDP (Ministerio de Hacienda, 2025; IMF, 2025).

Total external debt stood at approximately USD 254 billion (73% of GDP) at end-2023, concentrated in mining and energy (Banco Central de Chile, 2024; IMF, 2025). The current account has run deficits in most years since 2013, financed primarily by FDI. FX reserves of USD 45.6 billion cover over seven months of imports, providing a buffer against the country's main source of external volatility (IMF, 2025).

The Central Bank operates an inflation-targeting regime with a 3% target and 2–4% tolerance bands. The 2022–2024 tightening cycle (0.5% to 11.25%, then easing to 5.00%) demonstrated institutional credibility. The banking sector is well-capitalised by international standards, ranking among OECD-quality peers, but capital markets remain underdeveloped relative to other advanced economies, with limited equity market capitalisation and turnover.

Economic complexity and export concentration. Chile's ECI peaked at approximately -0.05 in the mid-2000s and has since declined to around -0.30 in 2022, ranking the country roughly 70th globally. Its export basket has become less diversified over time, as its manufacturing sector has declined from its peak in the early 1990s. Currently, services account for roughly 57% of GDP (2023) and manufacturing for around 10%.

Chile's products with revealed comparative advantage are clustered around copper, lithium, molybdenum, wood pulp, and fresh fruit. These products sit in the "periphery" of the product space (Hidalgo et al., 2007), where few nearby, higher-complexity products are available to branch into.

The products Chile has added to its export basket between 2009 and 2024 are somewhat more complex than its traditional exports, but remain resource-adjacent: chemical wood pulp, processed fats, and alloy steel, rather than the manufacturing or technology products that would shift the country's position in the product space. According to the Harvard Growth Lab (2025b), the diversification targets with the highest opportunity scores tend to be distant from Chile's current productive capabilities.

4.3.4 Human Capital

Chile's human capital reveals a quantity-quality gap. Average schooling of approximately 10 years places the country among the most educated in Latin America, yet educational outcomes lag: PISA mathematics scores sit 60 points below the OECD average, a shortfall possibly compounded by fewer teaching hours than peer countries (OECD, 2024a). R&D expenditure at 0.34% of GDP is roughly one-sixth of the 2022 OECD average, limiting the country's capacity to diversify and modernise its economy.

These shortcomings carry over to productivity. Worker output is approximately 50% below the OECD average (OECD, 2022), as workers tend to be highly productive only in the mining sectors. Antofagasta, the country's main mining region, records labour productivity 120% above Chile's national average (OECD, 2024b), while southern agricultural and service-oriented regions lag far behind.

4.3.5 Infrastructure and Technology

Chile's physical infrastructure is adequate by Latin American standards but still has clear bottlenecks to address, given its railroad infrastructure quality is ranked 70th out of 103 countries in the WEF Global Competitiveness Index (2019). The northern and southern rail systems operate on different gauges with no through connection, and most of the northern trunk line is unreliable (OECD, 2017b). Consequently, most mineral freight moves by road to port, raising transport costs. Port infrastructure quality is above the Latin American average but sits below the OECD median on the World Bank's Logistics Performance Index (2023).

Salinas (2021) estimates that closing Chile's infrastructure gap relative to comparable advanced economies could raise complex exports by roughly 23%, indicating that transport infrastructure acts as a direct constraint to its ability to diversify its export market.

4.3.6 Governance and Institutions

Chile ranks between the 84th and 91st global percentile across all six World Governance Indicators. Resource management has been institutionalised since the 1980s led by the state copper company CODELCO, while lithium extraction operates through time-limited licences with royalty obligations (OECD, 2022).

Weaknesses, however, persist at the operational level. Contract enforcement is slow and costly by OECD standards (World Bank, 2020). The failed constitutional processes of 2022 and 2023 have generated widespread political instability. Furthermore, gaps remain in fiscal transparency. The IMF's 2024 Article IV consultation flagged government debt items not fully captured by the headline fiscal balance and recommended improved disclosure.

4.3.7 Natural Resource Industry and Supply Chain

The mining sector comprises 363 producing facilities across 17 commodity groups, generating an estimated USD 62 billion in annual mineral value (2024). Copper accounts for $48.3 billion of this total, followed by lithium carbonate ($3.3B), gold ($2.2B), iodine ($1.7B), and molybdenum ($1.6B) (Figure 4.3). Production is heavily concentrated in three northern regions of Antofagasta, Tarapacá, and Atacama, which account for over 85% of output value (Figure 4.4). Antofagasta alone holds the largest share of mineral wealth while hosting just 2% of the population, anchored by Escondida, the world's largest copper mine with $11.7 billion in annual output (Figure 4.5).

Mining consumes 33% of national electricity and depends on desalination for water supply (ITA, 2025), creating upstream linkages that historically offer the most realistic diversification channel for resource economies (Gelb, 2010). A domestic equipment and services sector, estimated at USD 8–10 billion in annual revenue, has developed around these needs, covering transport, logistics, maintenance, engineering, and specialised inputs.

Forward linkages are more limited. Chile exports primarily in concentrate and refined form rather than fabricated products. Approximately 54% of copper exports go to China for downstream processing, while lithium carbonate is shipped to battery manufacturers in China, South Korea, and Japan (Figure 4.6). Despite a large amount of domestic processing, domestic value-added beyond refining remains modest. The National Lithium Strategy, announced in 2023, aims to capture more downstream processing domestically, but barriers to entry are high given fierce competition from established market leaders in East Asia.

Figure 4.3. Production value by mineral category, estimated 2024 (USD)

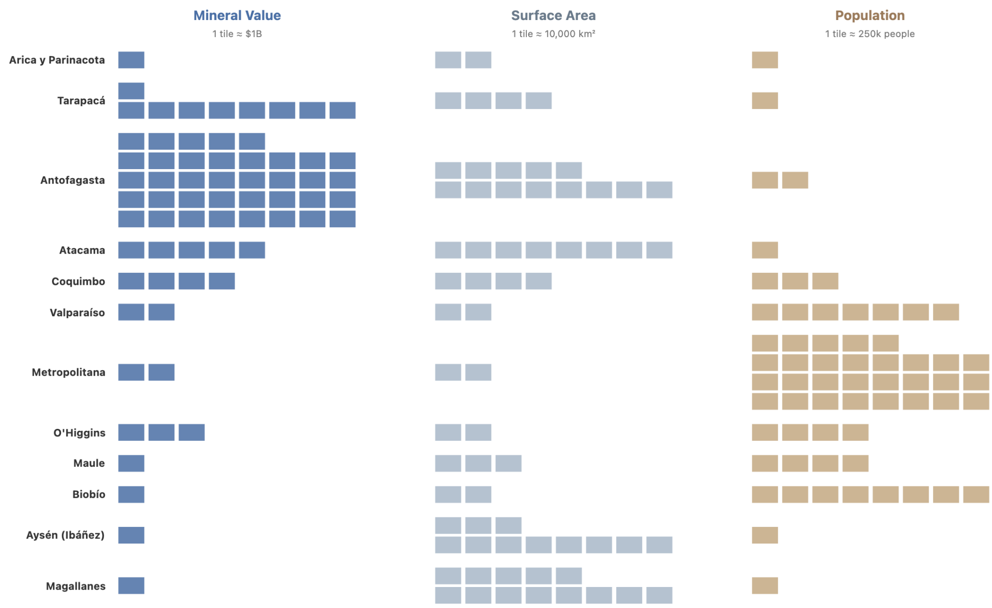

Figure 4.4. Regional distribution of mineral production value

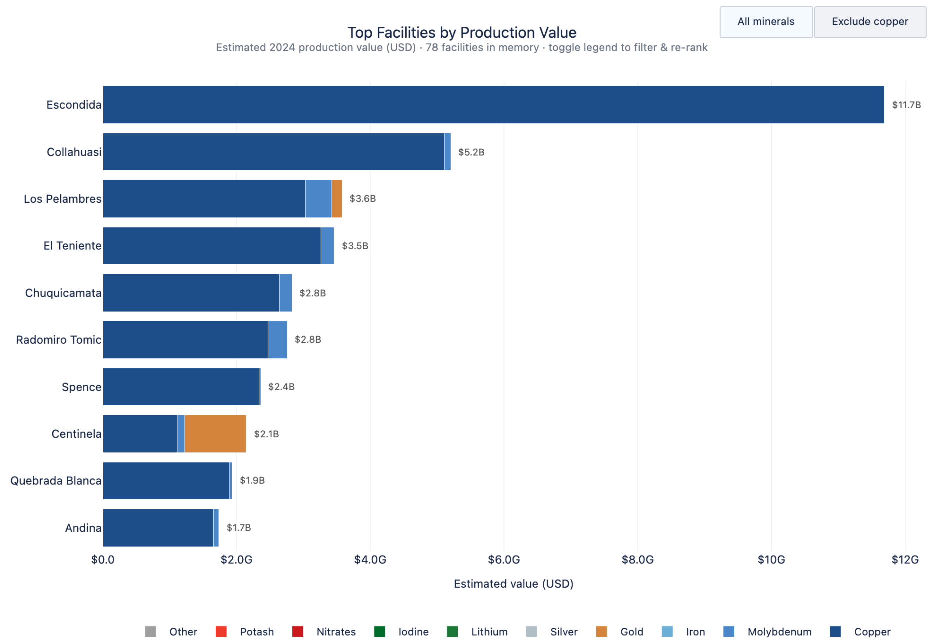

Figure 4.5. Top producing facilities by estimated annual output value (USD)

Figure 4.6. Mineral export destinations by value

4.3.8 Peer Comparison

Comparing Chile to other mineral producers illustrates how similar resource endowments can yield very different economic trajectories (Figure 4.7). Strong institutions explain the gap with countries like Zambia, but as the cases of Botswana and Norway show, governance alone does not produce diversification without sustained investment in human capital and R&D.

Zambia: copper dependence without institutional safeguards. Zambia is similarly copper-dependent to Chile (over 70% of exports), but at roughly one-eighth of Chile's GDP per capita (approximately USD 3,900 PPP). Its credit profile is, in nearly every aspect, the inverse of Chile's: the country defaulted on its Eurobond obligations in 2020 and only completed its debt restructuring in 2024. Frequent changes to the mining royalty regime, the absence of a sovereign wealth fund, and limited central bank independence have undermined long-term fiscal sustainability. While a direct comparison is complicated by differences in climate, geography, and history, the process of building a successful resource-intensive economy lies on institutional stability, fiscal discipline, and sustained investment in human capital.

Botswana: fiscal stability without structural transformation. Botswana used diamond revenues to rise from extreme poverty at independence to upper-middle-income status, maintaining fiscal surpluses through the Pula Fund (USD 4.5 billion at end-2023) and strong governance, scoring 43rd globally on Transparency International's CPI (2024) with uninterrupted multiparty elections since 1966. Yet diamonds still account for roughly 80% of merchandise exports, and Botswana's ECI remains among the lowest in the world. Both Chile and Botswana save during booms, spend countercyclically, and maintain fiscal discipline, but neither has truly solved the diversification problem. This suggests that strong fiscal and monetary institutions, while necessary to avoid the most severe forms of the resource curse, are not independently sufficient for long-term industrial upgrading.

Norway: what investment in human capital can achieve. Norway represents an aspirational peer to Chile, with considerably larger SWFs. Norway's Government Pension Fund Global (USD 1.6 trillion, 300% of GDP) dwarfs Chile's combined sovereign funds (under 5% of GDP), and R&D spending (2% of GDP vs. 0.34%) reflects a fundamentally different investment model. Norwegian universities and research institutes were integrated into the oil industry, producing engineering firms that later diversified into offshore wind and subsea technology (Norskpetroleum.no, 2024).

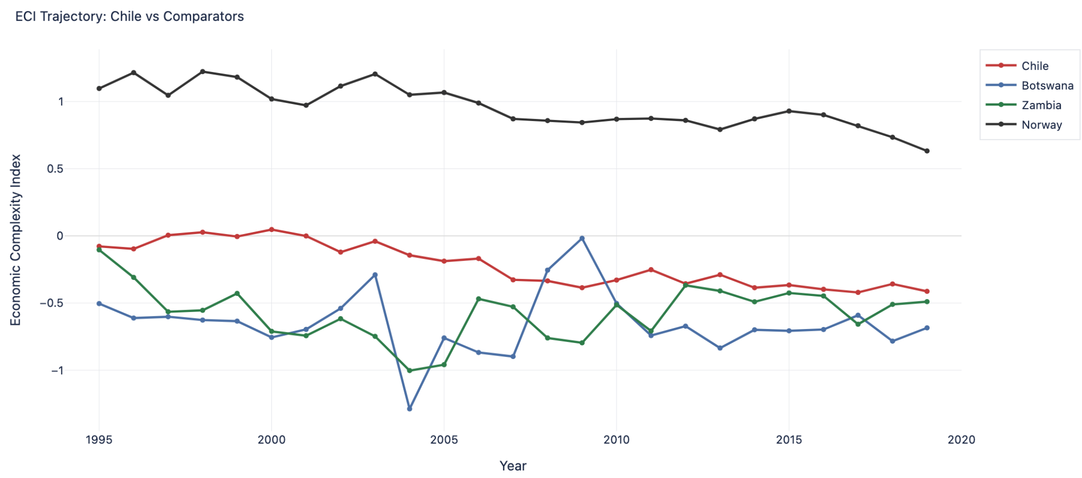

Figure 4.7. ECI trajectory: Chile and comparators

4.3.9 Policy Recommendations

Although Chile's institutional framework has shielded the economy from the worst forms of the resource curse, its impact on diversification remains limited: the ESSF is depleted, R&D spending is one-sixth of the OECD average, upstream service firms lack access to credit, and most mineral freight still moves by road through a corridor approaching capacity limits.

Rebuild the Economic and Social Stabilisation Fund. The ESSF fell from USD 20 billion (14% of GDP) in 2008 to approximately USD 5 billion (1.5%) by 2024, largely through pandemic-era drawdowns. Rebuilding the fund to 5–8% of GDP would help protect the investment programmes recommended below from being cut during the next commodity downturn.

Direct lithium revenues toward education quality and applied research. The econometric analysis finds the returns to educational investment are highest where extraction is most intensive. In Chile, that is Antofagasta, Tarapacá, and Atacama: the three northern regions that account for over 85% of mineral production value. Norway's petroleum sector illustrates the mechanism: research institutions such as SINTEF and the University of

Stavanger were operating in oil-producing regions (Ryggvik, 2015). Earmarking a defined share of lithium rents for applied research institutes and technical training in the northern mining corridor would encounter the interaction between education, research, and production.

Build the domestic mining supply chain through credit access and procurement policy. The upstream services sector is the most realistic near-term diversification channel. Procurement preferences for domestic suppliers (as in Norway, Australia, and Canada) and dedicated CORFO credit lines with risk-sharing facilities for firms lacking mining-scale collateral can direct capital where diversification is most likely to originate.

Invest in logistics infrastructure along the northern mining corridor. The northern mining regions depend on port facilities at Antofagasta and Mejillones that are approaching capacity limits, while the proposed lithium processing facilities will require additional energy, water, and transport connections. The northern and southern rail systems still operate on different gauges with no through connection, and most mineral freight moves by road. Investing in port expansion, rail standardisation, and energy connections along the northern corridor would do more for export competitiveness than any further tariff reduction.

Chapter 5

Final Remarks

This study examines how emerging market economies have struggled to translate natural resource extraction into higher value-added industrial activity, and which economic, institutional, and policy conditions explain cross-country divergence. It follows two complementary approaches, econometrics and machine learning methods, and analyses three case studies. The convergence of findings across these different approaches suggests the following conclusions.

Industrial upgrading in resource-rich economies is fundamentally a question of structural transformation. Three pillars emerge as the most robust determinants of economic complexity: domestic credit to the private sector (a proxy for financial depth), infrastructure capacity, associated with access to electricity, and human capital.

The three case studies contextualise these findings across different institutional and developmental settings, illustrating which constraints are binding at different levels of economic development. The ROC represents a severe form of persistent underperformance, ranking amid the worst in ECI. The country's pro-cyclical fiscal expenditure, chronically volatile education investment and limited electricity access suppress the conditions for diversification. Conversely, Azerbaijan presents a revealing case in which GDP grew but ECI declined. This dynamic is consistent with Dutch Disease crowding out non-oil tradable sectors. This condition contrasts with Malaysia, a comparable hydrocarbon economy whose ECI increased, showing that divergent complexity trajectories among oil-rich peers are linked to policy decisions regarding the reinvestment of resource rents.

Chile offers a third and distinct configuration, in which strong institutions, a structural fiscal balance rule, and the highest sovereign credit rating in Latin America coexist with a declining ECI since the mid-2000s. Unlike the ROC or Azerbaijan, the binding constraint here is not governance or macroeconomic instability, but human capital quality. With R&D investment at roughly one-sixth of the OECD average, Chile's capacity to capture value beyond extraction remains limited. Together, the three cases show that the constraints on industrial upgrading shift as institutional and economic conditions improve, and that no single policy lever is sufficient across all settings.

The forward-looking projections to 2030 reinforce the interpretation of these findings. Countries such as Ecuador and Mongolia, where financial depth, human capital, and institutional quality are relatively strong for their current ECI levels, are positioned for continued complexity gains. Conversely, hydrocarbon-dominant economies such as Saudi Arabia and Libya face projected declines unless structural reforms are implemented to address the deficits the models identify.

The policy implications that emerge from this analysis are consistent and mutually reinforcing across the quantitative and case study evidence. Deepening financial systems and expanding access to credit for non-extractive sectors constitutes the most consistent policy lever. Investment in human capital must be counter-cyclical and structurally insulated from commodity price volatility. The ROC's experience demonstrates that procyclical education spending produces capability deficits, while unreliable energy infrastructure suppresses private investment. Azerbaijan's trajectory confirms that institutional reforms, particularly resource management transparency, are prerequisites for productive returns outside the extractive sector. Chile's case further illustrates that even with strong macroeconomic fundamentals, the transition to higher-value activities requires deliberate industrial policy paired with investment in education quality and applied research to bridge the gap between extraction and high-value manufacturing.

Taken together, the evidence points toward a consistent policy implication: the most effective pathway for resource-rich developing economies is not to abandon dependence on natural resources but to systematically transform resource rents into the structural foundations needed to sustain long-term productive diversification and value-added industries: financial development, human capital, and reliable infrastructure.

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Appendices

Appendix A

Data Sources

A.1 Economic Complexity Index

ECI is constructed from goods trade data at the HS4 level of product classification. Countries and products are arranged into a binary matrix indicating whether a country holds a revealed comparative advantage in each product, defined as a share of that product in the country's exports exceeding its share in world exports. From this matrix, two quantities are derived: the diversity of each country's export basket and the ubiquity of each product it exports. These are then refined iteratively to account not only for how many products a country exports competitively, but for how rare those products are across other exporters. The resulting index summarises an economy's productive sophistication and serves as the proxy for value chain upgrading capacity throughout this analysis.

Employing ECI as an outcome carries limitations. Because it is derived from trade data, it excludes value-added activity that is domestically consumed, likely understating the degree of productive upgrading. It also omits the services sector. Finally, ECI measures accumulated productive know-how rather than the dollar value generated by value-added industries. Despite these limitations, ECI aligns well with the analytical framework of this study and offers wide availability across countries and years, covering the full panel from 1995 to 2019 with minimal gaps.

A.2 World Bank Data

The World Development Indicators (WDI) database provides a comprehensive set of statistics tracking economic, social, and environmental performance across countries worldwide (World Bank, 2026). We draw 26 WDI variables covering natural resource rents, the sectoral composition of GDP, savings and resource depletion indicators, employment shares, infrastructure proxies, financial indicators, demographics, and trade openness.

A.3 IMF WEO

We supplement the World Bank data with macroeconomic indicators from the IMF's World Economic Outlook: GDP per capita at constant prices in purchasing power parity, general government revenue as a share of GDP, general government net debt as a share of GDP, and the general government structural fiscal balance as a share of potential GDP (IMF, 2025).

A.4 IMF Investment and Capital Stock Dataset

We extract gross fixed capital formation (GFCF) as a share of GDP at constant prices and general government primary net lending as a share of GDP. These complement the WEO fiscal indicators by capturing the investment dimension of macroeconomic policy.

A.5 V-Democracy

We include indicators from the Varieties of Democracy (V-Dem) dataset, which provides expert-coded measures of political institutions, governance, and democracy. These indicators allow us to account for the role of governance and institutional capacity in the relationship between natural resources and value chain upgrading.

A.6 Penn World Table

We draw on data from the Penn World Table version 11.0 (PWT 11.0), specifically human capital variables, the capital depreciation rate and capital stock of countries (Feenstra, Inklaar, and Timmer, 2015).

A.7 CEPII

We include a binary indicator identifying whether a country is landlocked, sourced from the GeoDist database maintained by CEPII (Mayer and Zignago, 2011).

A.8 Natural Resources Stock and Production

We complement the rent-based proxies from the World Bank with direct measures of natural resource production quantities, reserves, and prices from the Energy Institute's Statistical Review of World Energy, Our World in Data, and the USGS Data Series 140 (Kelly and Matos, 2014).

A.8.1 Cross-Country Panel

Four primary sources are used. The Energy Institute's Statistical Review of World Energy provides annual production and consumption data for oil, natural gas, and coal, as well as production and reserve figures for 13 minerals, including cobalt, lithium, graphite, rare earths, copper, manganese, nickel, zinc, platinum group metals, bauxite, aluminium, tin, and vanadium (Energy Institute, 2024). We supplement these with mineral-level production and reserve series from Our World in Data (OWID), which covers over 20 minerals and draws on the British Geological Survey, the USGS, and other national geological surveys. Where available, we also incorporate mineral unit values from the U.S. Geological Survey Data Series 140 (USGS DS140), which provides historical price and value data for a range of non-fuel minerals (Kelly and Matos, 2014).

Price data, used to convert physical production quantities into dollar-denominated values, are drawn from a consolidated price file covering 19 minerals plus oil, coal, and natural gas over the period 1990 to 2024. Additional benchmark prices (Brent crude, Henry Hub natural gas, TTF, JKM, and the PCOALAU thermal coal index) serve as fallbacks when primary series are unavailable. The resulting dataset provides country-year estimates of production value for hydrocarbons and metals, which we merge into the master panel as additional measures of resource endowment.

A.8.2 Chile Facility-Level Supply Chain

For the Chile case study, we move beyond aggregate country-year figures to construct a facility-level supply chain covering all minerals Chile produces. The mineral facility inventory, recording geographic coordinates, facility type, operator, and resource and reserve tonnages for over 460 sites, is sourced from Sernageomin (Chile's national geological survey), supplemented by the USGS Mineral Resources Data System. Mine-to-plant linkages from the same sources allow us to trace physical flows from extraction through concentration, smelting, and refining.

Annual production volumes for copper (by company and by region), molybdenum (by mine), and over 25 non-metallic minerals (by region) are taken from the COCHILCO Anuario Estadístico (COCHILCO, 2024), which also provides commodity prices from the London Metal Exchange, exchange-traded benchmarks, and midpoint estimates for non-exchange minerals such as lithium carbonate and iodine. For minerals not covered in the Anuario price tables, we use implied FOB unit values derived from COCHILCO export value and volume series, or benchmark prices from the USGS Mineral Commodity Summaries and the Platts IODEX iron ore index.

Export flows are constructed from two trade sources at different levels of granularity. Shipment-level customs records from the Servicio Nacional de Aduanas provide individual export transactions with port of departure, destination country, HS product code, and FOB value, allowing us to observe port-level commodity routing. We cross-validate these against UN Comtrade bilateral trade data at the HS6 level and use Comtrade as the primary source of destination shares for minerals where customs coverage is incomplete. The combination of facility-level production, processing linkages, and port-level trade data produces a directed network of approximately 460 facilities and 1,900 edges across 17 commodity groups, which we use to characterise the structure and concentration of Chile's mineral value chain from mine to export destination.

Appendix B

Processing Data

B.1 Countries Excluded Due to Data Availability

Table B.1 lists the countries excluded from the sample and the rationale for each exclusion.

Table B.1. Countries excluded from the sample

Country	Code	Reason
French Guiana	GUF	Little economic data; minor gold producer
North Korea	PRK	Extensive missing economic data; reliability concerns
Taiwan	TWN	Limited natural resources, poor data coverage
New Caledonia	NCL	French territory; governance structure complicates analysis
South Sudan	SSD	Extensive missing economic data
Cuba	CUB	Extensive missing economic data
Eritrea	ERI	Poor economic data coverage
Timor-Leste	TLS	Highly local economy; poor economic data
Afghanistan	AFG	Key variables missing
Turkmenistan	TKM	Key economic variables missing completely
Syria	SYR	Insufficient variables with decent coverage
Montenegro	MNE	ECI and human capital variables missing
Cambodia	KHM	Only features for 1 year; minor gold production

B.2 Imputation

Linear interpolation is a numerical method used to estimate missing values between two observed data points by assuming that the relationship between them is linear. In simple terms, it draws a straight line between the known values and uses that line to approximate the missing observations (Burden and Faires, 2011). If a variable is observed at time t_0 with value y_0 and at time t_1 with value y_1, any missing value between those points is estimated proportionally based on its relative position in time.

Formally, the interpolated value at time t (where $t_0 < t < t_1$) is calculated as:

$$y_t = y_0 + \frac{(y_1 - y_0)}{(t_1 - t_0)} \times (t - t_0) \tag{B.1}$$

This formula distributes the total change between y_0 and y_1 evenly across the time interval. The method therefore assumes a constant rate of change between the two observed points. Where multiple consecutive values are missing, each is filled along the same linear trajectory. In our panel context, interpolation is performed strictly within each country's own time series.

Extrapolation applies the same linear logic beyond the first or last observed value, extending the trend outward to fill leading or trailing gaps. To limit the risk of projecting short-run patterns over extended periods where the true trajectory is unknown, extrapolation is capped at three years beyond the nearest observed value. Gaps exceeding this window are left for the KNN pass described below. This approach performs well for variables that change gradually over time, such as population, capital stock, or human capital. It may understate the effect of structural breaks or abrupt policy shifts.

B.3 K-Nearest Neighbours Imputation

Where a country has no observed value for a given variable across the entire sample period, interpolation cannot proceed. These cells are filled in a second pass using K-Nearest Neighbours imputation (KNN, $k = 5$, distance-weighted). For each missing cell, KNN identifies the five most similar country-year observations in the dataset, measured by Euclidean distance across all non-missing variables, and fills the cell with a distance-weighted average of those five values. Unlike interpolation, KNN draws on information from other countries and, because similarity is computed at the country-year level, across time periods simultaneously. To assess imputation accuracy, a leave-out validation test masks five percent of known values at random, applies KNN to fill them, and compares the imputed values to the withheld observations.

The median prediction error across all variables is 5.24%. Performance is weaker for volatile time-series: inflation (31.1%), real interest rates (36.4%), and mineral rents (44.4%). These variables should be interpreted with additional caution in the regression output. Within the analytical sample of 54 countries, 5,323 cells are filled by KNN. Twenty-five countries have more than ten percent of their total data cells imputed by this procedure rather than observed or interpolated, indicating a heavier dependence on cross-country assumptions for those cases.

Appendix C

Clustering Techniques

C.1 Principal Component Analysis

Principal Component Analysis (PCA) is a dimensionality reduction technique that transforms a set of potentially correlated variables into a smaller set of uncorrelated variables called principal components (PCs). Each principal component is a linear combination of the original variables, constructed to capture the maximum variance remaining in the data.

Given a data matrix **X** of dimension $n \times p$ (where n is the number of observations and p is the number of variables), PCA seeks to find an orthogonal transformation that projects the data onto a new coordinate system. The first principal component has the largest variance, the second principal component has the second largest variance (orthogonal to the first), and so on (Jolliffe, 2002). The analysis involves the following steps:

Step 1: Standardization. Since variables are measured in different units, we first standardize the data matrix so each variable has mean zero and variance one: $Z_{ij} = \frac{X_{ij}-\mu_j}{\sigma_j}$

Step 2: Covariance Matrix. $\Sigma = \frac{1}{n-1} Z^T Z$

Step 3: Eigendecomposition. $\Sigma \mathbf{v}_k = \lambda_k \mathbf{v}_k$

where λ_k are the eigenvalues (representing variance explained by each component) and $\mathbf{v}_k$ are the corresponding eigenvectors (the principal component directions). The eigenvalues are ordered such that $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_p$.

Step 4: Principal Component Scores. $PC = ZV$

where V is the matrix of eigenvectors. Each column of PC represents a principal component.

C.1.1 Factor Loadings

In our analysis, we apply this procedure to 21 resource production variables measured across the sample countries. The first two principal components together account for 69.5% of total variance in the standardised data. PC1 explains 51.8% and is driven primarily by oil and natural gas production; PC2 explains an additional 17.7% and captures variation in mining activity, particularly copper, gold, and coal.

Figure C.1 displays the full set of factor loadings, showing how each of the 21 resource variables contributes to the two retained components. Oil and natural gas carry the highest loadings on PC1, while copper, gold, and coal dominate PC2, with moderate contributions from iron ore, aluminium, and silver. Most smaller minerals (cobalt, lead, tin, lithium, among others) load weakly on both components, reflecting either limited production volumes or concentration in too few countries to drive cross-country variation.

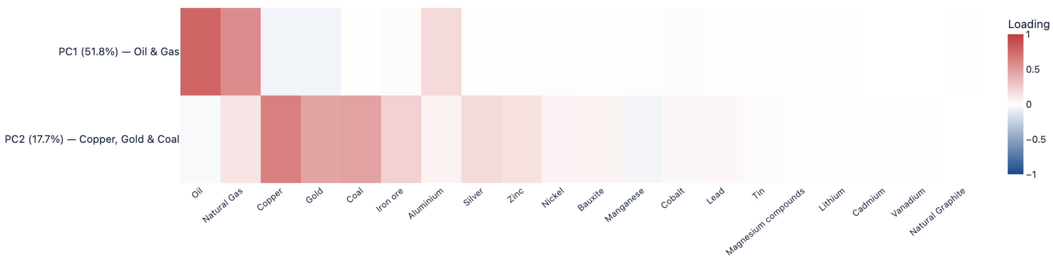

Figure C.1. PCA loadings heatmap: resource production profiles

C.2 K-means Clustering Algorithm

K-means is an unsupervised clustering algorithm that divides n observations into k clusters by minimizing within-cluster variance (MacQueen, 1967). We apply K-means to the first two principal components to identify distinct groups of countries.

K-means aims to minimise the within-cluster sum of squares (WCSS), defined as:

$$WCSS = \sum_{j=1}^{k} \sum_{i \in C_j} \|x_i - \mu_j\|^2 \tag{C.1}$$

where C_j denotes the set of observations in cluster j, x_i is the ith observation, and μ_j is the centroid (mean) of cluster j. This is done in the following iterative steps:

1. Initialization: Randomly select k initial cluster centroids from the data points.
2. Assignment: Assign each observation to the nearest centroid (using Euclidean distance).
3. Update: Recalculate each centroid as the mean of all observations assigned to that cluster.
4. Convergence: Repeat steps 2–3 until cluster assignments no longer change or the maximum number of iterations is reached.

C.3 Cluster Robustness

The baseline clustering in the main text applies Principal Component Analysis (PCA) followed by K-Means ($k = 5$) to 21 log-transformed, per-capita resource production variables for the 1995 cross-section of 54 countries. This appendix performs several checks to assess how sensitive the assignments are to modelling choices.

C.3.1 Choice of k

The silhouette score, which measures how similar each country is to its own cluster relative to the nearest neighbouring cluster (values closer to 1 indicate tighter, better-separated groups), is used to restrict the size of our clusters. The ideal choice is somewhere between 3 and 6 (Figure C.2).

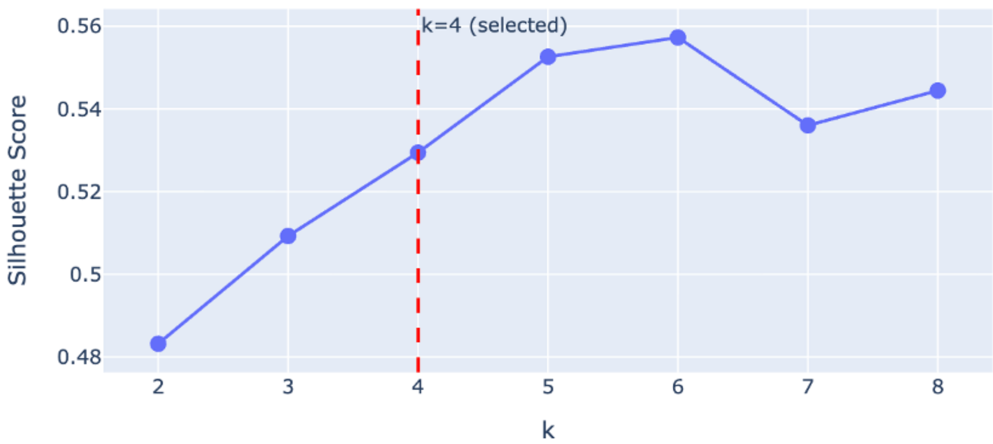

Figure C.2. Silhouette score across clusters

C.3.2 Alternative Cluster Specifications

At $k = 3$, the original Petrostates and Oil Exporters merge into a single hydrocarbon cluster, losing the distinction between high-income Gulf producers and lower-income oil-dependent economies.

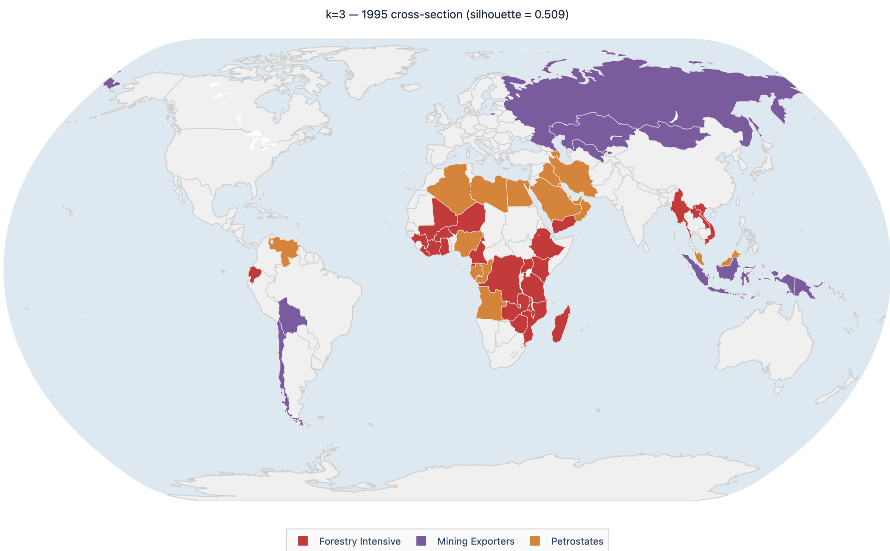

Figure C.3. Cluster map with $k = 3$

At $k = 4$, the original Mining Exporters and Diversified Producers groups collapse into a single "Major Producers" cluster. This absorbs Chile, Mongolia, Kazakhstan, Russia, Bolivia, Indonesia, and Papua New Guinea into one group, uniting countries whose per-capita production is dominated by minerals and those with mixed hydrocarbon-mineral endowments.

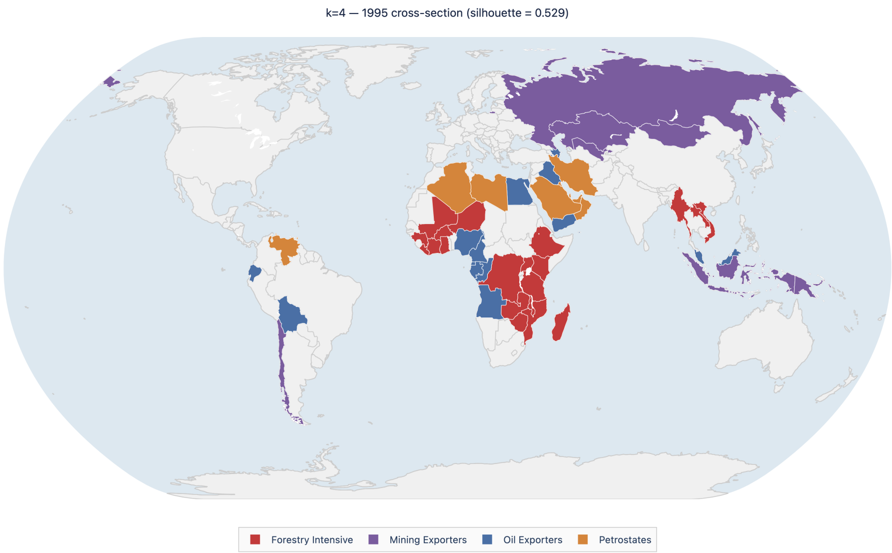

Figure C.4. Cluster map with $k = 4$

At $k = 6$, the Forestry Intensive group splits further, but the additional cluster contains only two or three countries and does not correspond to a clearly interpretable resource profile. An "Oil & Minerals" cluster also appears, separating a handful of mixed-portfolio economies that at $k = 5$ are distributed between Oil Exporters and Diversified Producers.

The five-cluster solution ($k = 5$) is retained because it yields the most interpretable separation along both the hydrocarbon and mineral dimensions without producing clusters that are too small or too difficult to interpret.

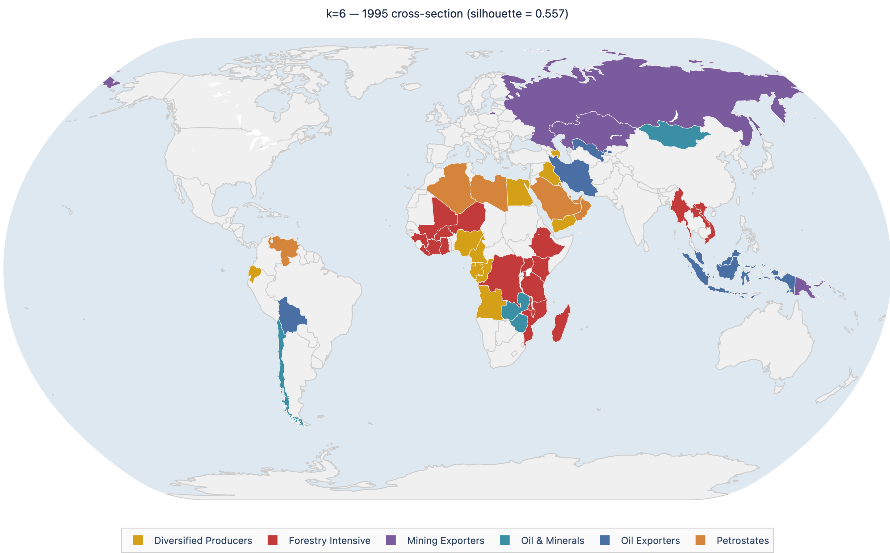

Figure C.5. Cluster map with $k = 6$

C.3.3 Alternative Specifications

Three robustness checks vary one feature of the baseline pipeline at a time while holding others constant. Agreement with the baseline is measured by the Adjusted Rand Index (ARI), which compares two partitions on a scale from 0 (random agreement) to 1 (identical assignments) and corrects for chance.

Robustness A: Merging mineral commodities. All non-hydrocarbon mineral resources are collapsed into a single "Minerals" column, retaining four features: Oil, Natural Gas, Coal, and Minerals. PCA (2 components) and K-Means are re-run on this aggregated feature set, with $k = 5$. The ARI between this specification and the baseline is 0.70, indicating substantial but not perfect agreement. The Petrostates and Oil Exporters clusters remain almost identical to the baseline, confirming that the hydrocarbon dimension is stable across both feature sets and cluster counts. The main source of divergence is in the treatment of mineral-dependent economies. In the baseline 21-feature model, PCA can distinguish copper-dominant profiles from gold-dominant or diversified-mineral profiles, which separates countries like Chile, Zambia, Mongolia, and Ghana from the low-production group.

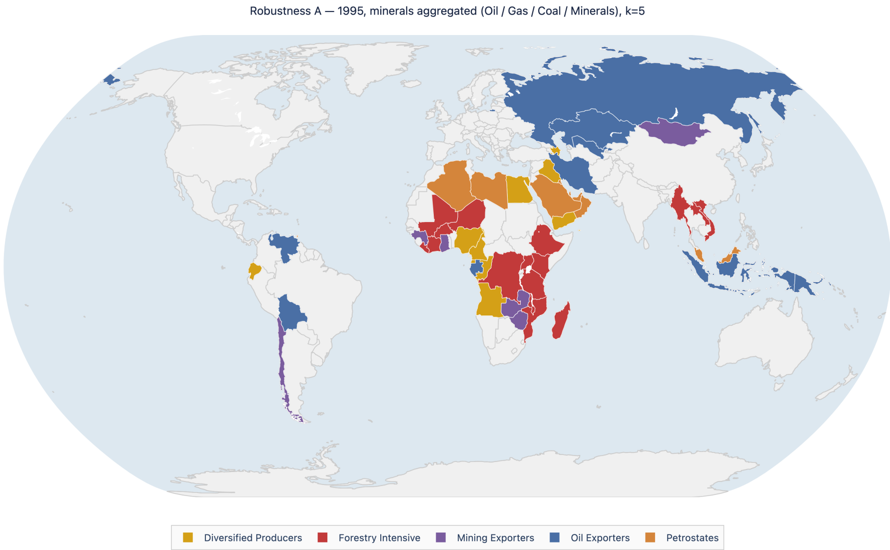

Figure C.6. Robustness A: Merged mineral commodities

Robustness B: Three principal components. Retaining three PCA components instead of two tests whether the reduction in information decreases the potential interpretability of the model and its ability to create meaningful clusters. The third component adds 11.4% of explained variance, bringing the total to approximately 81%. K-Means ($k = 5$) is applied in the three-dimensional PCA space. The ARI of 0.81 is the highest across all checks, suggesting that the third component adds limited power for clustering purposes. Most countries end up in the same group regardless of whether the clustering is performed in two or three dimensions, with few exceptions.

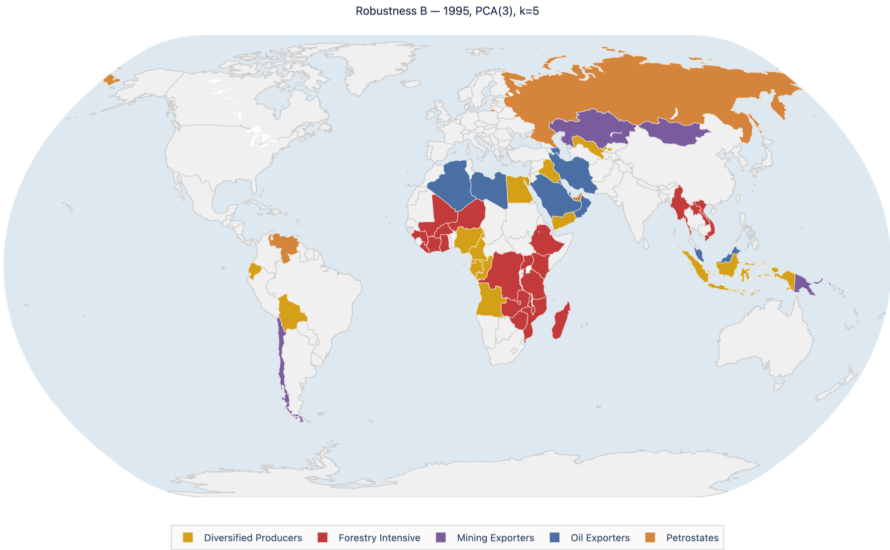

Figure C.7. Robustness B: Three principal components

Robustness C: Clustering on 2019-level production. Replacing the 1995 snapshot with 2019 data tests temporal stability while keeping the rest of the process identical (21 features, PCA(2), $k = 5$). The ARI of 0.67, the lowest of the three checks, reflects concrete changes in production profiles over 24 years. Azerbaijan and Iraq, which scaled oil production substantially between 1995 and 2019, move from Oil Exporters into the Petrostate group. Kazakhstan and Russia, where mineral extraction expanded relative to hydrocarbons, shift into Mining Exporters. Ghana, Myanmar, and Vietnam leave the Forestry Intensive group as their extraction portfolios broadened.

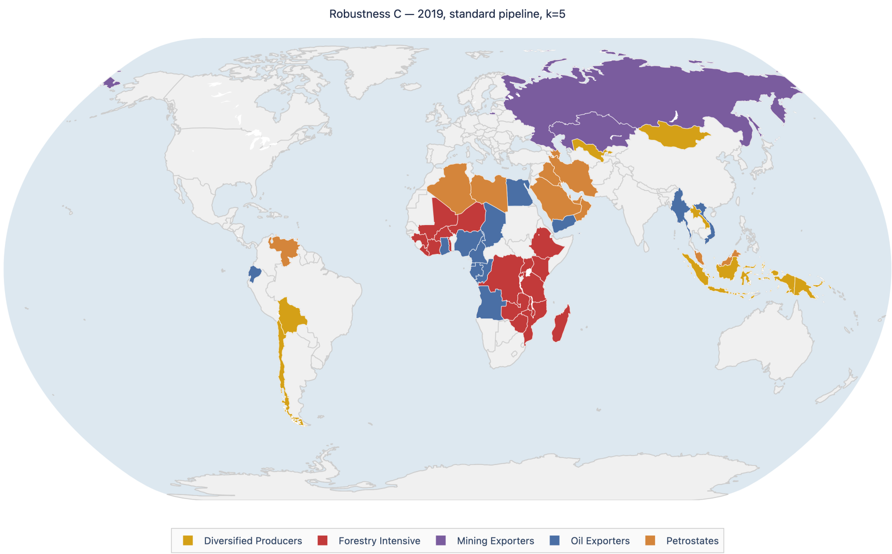

Figure C.8. Robustness C: 2019-level production

Country-level stability. Figure C.9 presents a heatmap of cluster assignments across all four specifications (baseline plus three robustness checks), with countries sorted by the number of label changes.

Fourteen countries (27%) receive the same label in every specification. These perfectly stable members are almost entirely drawn from the Forestry Intensive group (Burkina Faso, DRC, Côte d'Ivoire, Ethiopia, Kenya, Liberia, Madagascar, Malawi, Mali, Mozambique, Niger, Tanzania, Uganda) plus the United Arab Emirates. The Petrostates cluster is nearly stable: Kuwait, Qatar, Saudi Arabia, Libya, Oman, and Trinidad and Tobago retain their baseline label in all checks except Robustness B, where the additional dimension shifts them to Oil Exporters.

The least stable countries are those with mixed hydrocarbon-mineral portfolios. Bolivia, Indonesia, Kazakhstan, Papua New Guinea, Russia, Uzbekistan, and Iran shift label in two or more specifications. This boundary sensitivity is itself a substantive finding: these are the economies whose resource composition is least well summarised by a single cluster.

Cluster Label Assignments: Baseline vs Robustness Checks

	Baseline (k=5)	Rob A: Minerals merged	Rob B: PCA(3)	Rob C: 2019
Burkina Faso	Forestry Intensive	Forestry Intensive	Forestry Intensive	Forestry Intensive
Congo, Dem. Rep.	Forestry Intensive	Forestry Intensive	Forestry Intensive	Forestry Intensive
Cote d'Ivoire	Forestry Intensive	Forestry Intensive	Forestry Intensive	Forestry Intensive
Ethiopia	Forestry Intensive	Forestry Intensive	Forestry Intensive	Forestry Intensive
Kenya	Forestry Intensive	Forestry Intensive	Forestry Intensive	Forestry Intensive
Liberia	Forestry Intensive	Forestry Intensive	Forestry Intensive	Forestry Intensive
Madagascar	Forestry Intensive	Forestry Intensive	Forestry Intensive	Forestry Intensive
Malawi	Forestry Intensive	Forestry Intensive	Forestry Intensive	Forestry Intensive
Mali	Forestry Intensive	Forestry Intensive	Forestry Intensive	Forestry Intensive
Mozambique	Forestry Intensive	Forestry Intensive	Forestry Intensive	Forestry Intensive
Niger	Forestry Intensive	Forestry Intensive	Forestry Intensive	Forestry Intensive
Tanzania	Forestry Intensive	Forestry Intensive	Forestry Intensive	Forestry Intensive
Uganda	Forestry Intensive	Forestry Intensive	Forestry Intensive	Forestry Intensive
United Arab Emirates	Petrostates	Petrostates	Petrostates	Petrostates
Angola	Diversified Producers	Diversified Producers	Diversified Producers	Oil Exporters
Cameroon	Diversified Producers	Diversified Producers	Diversified Producers	Oil Exporters
Congo, Rep.	Diversified Producers	Diversified Producers	Diversified Producers	Oil Exporters
Ecuador	Diversified Producers	Diversified Producers	Diversified Producers	Oil Exporters
Egypt, Arab Rep.	Diversified Producers	Diversified Producers	Diversified Producers	Oil Exporters
Equatorial Guinea	Diversified Producers	Diversified Producers	Diversified Producers	Oil Exporters
Iraq	Diversified Producers	Diversified Producers	Diversified Producers	Petrostates
Nigeria	Diversified Producers	Diversified Producers	Diversified Producers	Oil Exporters
Yemen, Rep.	Diversified Producers	Diversified Producers	Diversified Producers	Oil Exporters
Guinea	Forestry Intensive	Mining Exporters	Forestry Intensive	Forestry Intensive
Lao PDR	Forestry Intensive	Forestry Intensive	Forestry Intensive	Diversified Producers
Myanmar	Forestry Intensive	Forestry Intensive	Forestry Intensive	Oil Exporters
Viet Nam	Forestry Intensive	Forestry Intensive	Forestry Intensive	Oil Exporters
Zambia	Forestry Intensive	Mining Exporters	Forestry Intensive	Forestry Intensive
Zimbabwe	Forestry Intensive	Mining Exporters	Forestry Intensive	Forestry Intensive
Chile	Mining Exporters	Mining Exporters	Mining Exporters	Diversified Producers
Mongolia	Mining Exporters	Mining Exporters	Mining Exporters	Diversified Producers
Algeria	Petrostates	Petrostates	Oil Exporters	Petrostates
Bahrain	Petrostates	Oil Exporters	Petrostates	Petrostates
Kuwait	Petrostates	Petrostates	Oil Exporters	Petrostates
Libya	Petrostates	Petrostates	Oil Exporters	Petrostates
Malaysia	Petrostates	Petrostates	Oil Exporters	Petrostates
Oman	Petrostates	Petrostates	Oil Exporters	Petrostates
Qatar	Petrostates	Petrostates	Oil Exporters	Petrostates
Saudi Arabia	Petrostates	Petrostates	Oil Exporters	Petrostates
Trinidad and Tobago	Petrostates	Petrostates	Oil Exporters	Petrostates
Venezuela, RB	Petrostates	Oil Exporters	Petrostates	Petrostates
Azerbaijan	Diversified Producers	Diversified Producers	Oil Exporters	Petrostates
Gabon	Diversified Producers	Oil Exporters	Diversified Producers	Oil Exporters
Ghana	Forestry Intensive	Mining Exporters	Forestry Intensive	Oil Exporters
Bolivia	Oil Exporters	Oil Exporters	Diversified Producers	Diversified Producers
Indonesia	Oil Exporters	Oil Exporters	Diversified Producers	Diversified Producers
Kazakhstan	Oil Exporters	Oil Exporters	Mining Exporters	Mining Exporters
Papua New Guinea	Oil Exporters	Oil Exporters	Mining Exporters	Diversified Producers
Russian Federation	Oil Exporters	Oil Exporters	Petrostates	Mining Exporters
Uzbekistan	Oil Exporters	Oil Exporters	Diversified Producers	Diversified Producers
Iran, Islamic Rep.	Petrostates	Oil Exporters	Oil Exporters	Petrostates

Figure C.9. Cluster assignment stability across specifications

Implications for the main analysis. The clustering is used in the main text as a descriptive device to structure the case-study discussion, not as an input to the econometric or machine-learning models. The hydrocarbon clusters and increased interpretability of the mineral group support the use of the five-cluster baseline method, though each of the three specifications is defensible. The sensitivity of boundary cases to feature aggregation and time period is a limitation of this type of unsupervised classification and a reason why it is not included as a predictor in the econometric or ML specifications.

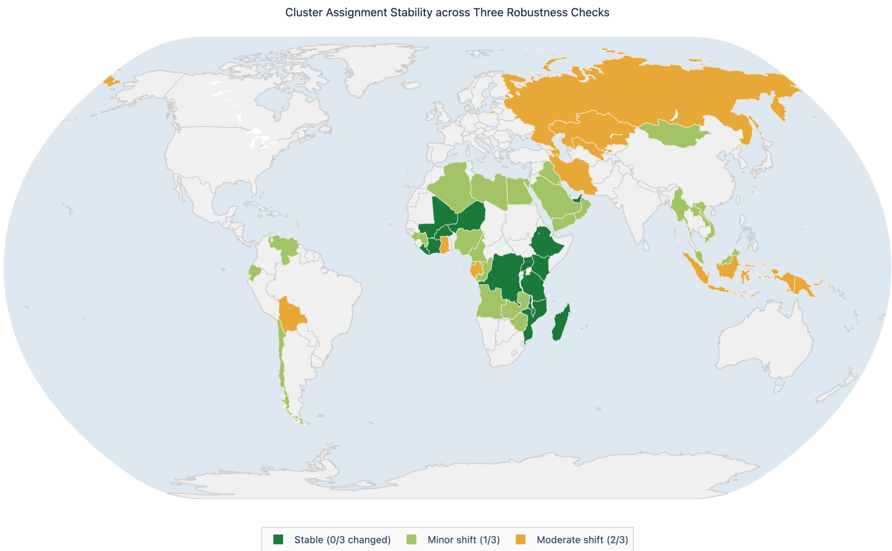

Figure C.10. Country-level cluster stability map

Appendix D

Regression Approach

D.1 Change in ECI as Dependent Variable

Given the high persistence of ECI, we ran a regression with the change in ECI as the dependent variable as an additional robustness check. By running the regression on the ECI change we eliminate the ECI dependency as the change does not depend on the current ECI level. However, to mitigate a possible correlation between the change and the ECI level, we added an error correction variable. The variable functions as an error-correction mechanism, capturing mean reversion toward the long-run equilibrium. The results are robust to this specification (Figure D.1).

Economic Complexity in Resource-Rich Countries — Error-Correction Specifications

	Dependent variable: ΔECI	
	(1)	**(2)**
Human Capital Index (log)	0.286***	0.276***
	(0.106)	(0.107)
Gross Fixed Capital Formation (log)	0.038***	0.034***
	(0.010)	(0.012)
Production Value (log)	-0.008	-0.007
	(0.008)	(0.008)
Political Stability	0.013	0.016
	(0.019)	(0.016)
Rule of Law	0.067	0.066
	(0.046)	(0.046)
Trade Openness (% of GDP)	0.000	0.000
	(0.001)	(0.001)
Forestry Rents (% of GDP)	-0.005	-0.004
	(0.004)	(0.004)
Interaction terms		
HCI × Production (log)	0.035	0.034
	(0.029)	(0.030)
GFCF × Production (log)	0.007	-0.002
	(0.006)	(0.006)
HCI × Forestry Rents	-0.023	-0.022
	(0.017)	(0.018)
GFCF × Forestry Rents	0.006*	0.001
	(0.004)	(0.003)
Error-correction term		
ECI (t−1) [error-correction]	-0.189***	-0.187***
	(0.044)	(0.044)
Constant	-0.558***	-0.544***
	(0.136)	(0.150)
Observations	1,270	1,270
Countries	54	54
R^2	0.098	0.095
Adjusted R^2	0.089	0.086
Clustered SE	Country	

Note: *p<0.1; **p<0.05; ***p<0.01. Clustered standard errors by country in parentheses. Dependent variable is the first difference of ECI ($\Delta ECI = ECI_t - ECI_{t-1}$). ECI(t−1) enters as an error-correction term capturing mean reversion. Interaction terms computed on mean-centered variables. Model (1): contemporaneous regressors. Model (2): all regressors lagged one period (t−1) to address endogeneity.

Figure D.1. Regression results: change in ECI as dependent variable

Appendix E

Modelling Details

E.1 Regularization Techniques

E.1.1 LASSO Regression

LASSO adds an L1 penalty to the least squares objective:

$$\min_{\beta} \left\{ \frac{1}{2n} \sum_{i=1}^{n} (y_i - x_i^T \beta)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\} \tag{E.1}$$

E.1.2 Ridge Regression

Ridge regression adds an L2 penalty:

$$\min_{\beta} \left\{ \frac{1}{2n} \sum_{i=1}^{n} (y_i - x_i^T \beta)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right\} \tag{E.2}$$

E.1.3 Elastic Net

Elastic Net combines L1 and L2 penalties:

$$\min_{\beta} \left\{ \frac{1}{2n} \sum_{i=1}^{n} (y_i - x_i^T \beta)^2 + \lambda \left(\alpha \sum_{j=1}^{p} |\beta_j| + \frac{1-\alpha}{2} \sum_{j=1}^{p} \beta_j^2 \right) \right\} \tag{E.3}$$

E.1.4 Random Forest

Random Forest (RF) is a non-parametric ensemble learning method (Breiman, 2001) that constructs a large number of decision trees on bootstrap samples drawn with replacement from the training data, and aggregating their predictions by averaging. Each bootstrap sample contains approximately 63.2% of the original observations; the remaining out-of-bag (OOB) observations are used to obtain an unbiased internal estimate of generalization error without requiring a separate validation set. This bagging procedure substantially reduces the variance that characterizes individual decision trees while leaving bias largely unchanged.

A key distinction from standard bagging is the introduction of random feature subsampling at each node split. Rather than evaluating all p available predictors, RF randomly selects a subset of $m < p$ candidates at each node, with m typically set to $p/3$ for regression tasks and tuned via cross-validation. This additional randomization further decorrelates the individual trees: even when a small number of strong predictors dominate the data, not every tree is built around those predictors, producing a more diverse and robust ensemble. We set $B = 500$ trees and tune m via cross-validation.

Random Forest produces two standard variable importance measures. The Mean Decrease in Impurity (MDI) accumulates the total reduction in the residual sum of squares attributable to splits on each predictor, averaged across all trees. Permutation importance (MDA) measures the increase in OOB prediction error when the values of a given predictor are randomly permuted, providing a more reliable, model-agnostic importance score that is not sensitive to predictor scale or cardinality.

Despite its flexibility and predictive power, Random Forest has limitations that justify its use as a robustness check rather than a primary estimator. Most importantly, it does not produce coefficient estimates: while variable importance scores indicate which predictors matter, they convey neither the direction nor the magnitude of the effect, limiting economic interpretability. As an ensemble of hundreds of trees, the model also lacks the transparency of a single linear equation, making results harder to communicate in a policy context.

E.2 Variance Inflation Factor (VIF)

Multicollinearity, when independent variables are moderately or highly correlated with one another, inflates the variance of the OLS estimates, making them unstable and difficult to interpret. In a regularization context, the degree of multicollinearity is equally consequential: it determines how each penalty structure interacts with the correlation among predictors, influencing the stability of coefficient estimates and the reliability of variable selection.

The VIF for variable j is defined as:

$$VIF_j = \frac{1}{1 - R_j^2} \tag{E.4}$$

where R_j^2 is the R-squared obtained from regressing variable j on all remaining predictors. A VIF of 1 indicates no collinearity; values between 1 and 5 are generally considered acceptable; values between 5 and 10 signal moderate multicollinearity; and values above 10 are considered severe and potentially problematic (Hair et al., 2019; Kutner et al., 2004).

The results of the VIF analysis (Figure E.1) indicate that almost all variables fall well below the conventional threshold of 5 in both models. The highest value recorded is 8.5 for Access to Electricity, with Production Value hovering around 5. While this might suggest some degree of multicollinearity, our choice of testing multiple ML specifications ensures that the aggregate performance of the three penalised methods will largely not be driven by collinearity.

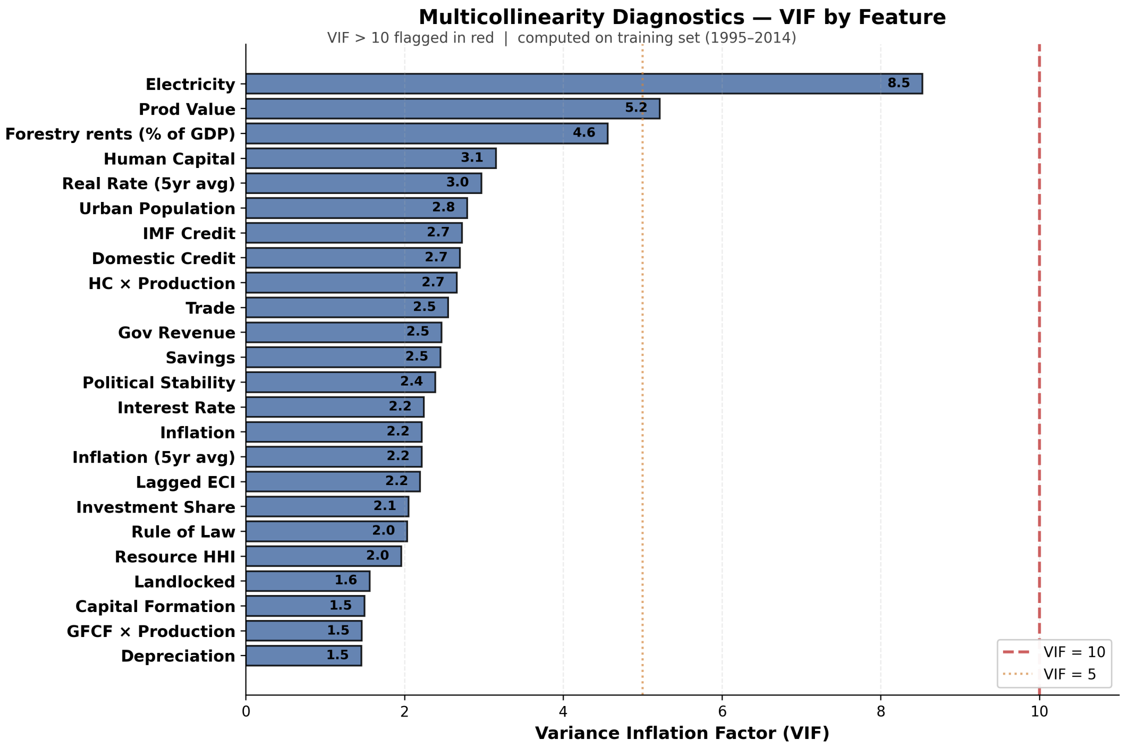

Figure E.1. Multicollinearity diagnostics: VIF by feature

Appendix F

Model Extensions

F.1 Bootstrap Model Coefficients

To test how sensitive the results are to the composition of countries in the sample, we run a country-level block bootstrap with 200 iterations. In each iteration, 54 countries are drawn with replacement, a fresh panel is assembled with suffix codes for duplicated countries, with the full imputation being applied before refitting all models. This draws uncertainty from both sample composition and imputation into the reported distributions, capturing sources of variation that classical standard errors do not reflect.

The main OLS findings survive resampling. Human Capital (Log HCI) and Rule of Law are positive in both specifications with 100% sign stability, and their 95% bootstrap confidence intervals exclude zero. Lagged ECI is stable in sign across 95% of draws, though its bootstrap median falls below the original point estimate, indicating that its persistence is sensitive to which countries appear in each resampled panel. Among the ML models, Random Forest achieves the highest median test R^2 (0.776) with all three penalised linear models clustering tightly around 0.717 to 0.725. Full-sample R^2 values (0.85 to 0.88) sit above these medians, as expected given that bootstrap panels are on average smaller and contain duplicate country-year sequences. The feature importance rankings that emerge from the ML models are consistent with the OLS results.

Electricity Access dominates Random Forest importance, with a median Gini of approximately 0.20 and a wide 95% confidence interval reflecting instability in its exact magnitude across resampled panels. Human Capital and Rule of Law follow, while Domestic Credit and Urban Population also appear consistently in the top five. Lagged ECI, which was the strongest persistence term in OLS, ranks lower under Random Forest, suggesting that much of its predictive content is absorbed by the structural variables that the tree-based model can exploit non-linearly. LASSO selection frequencies tell a broadly similar story, with Human Capital, Electricity Access, and Domestic Credit being retained in at least 95% of bootstrap samples. Rolling macro controls fall below 50%, suggesting they contribute predictive value only under specific sample compositions. Across both modelling approaches, the same core variables, human capital, institutional quality, infrastructure access, and financial depth, emerge as the most stable predictors of economic complexity.

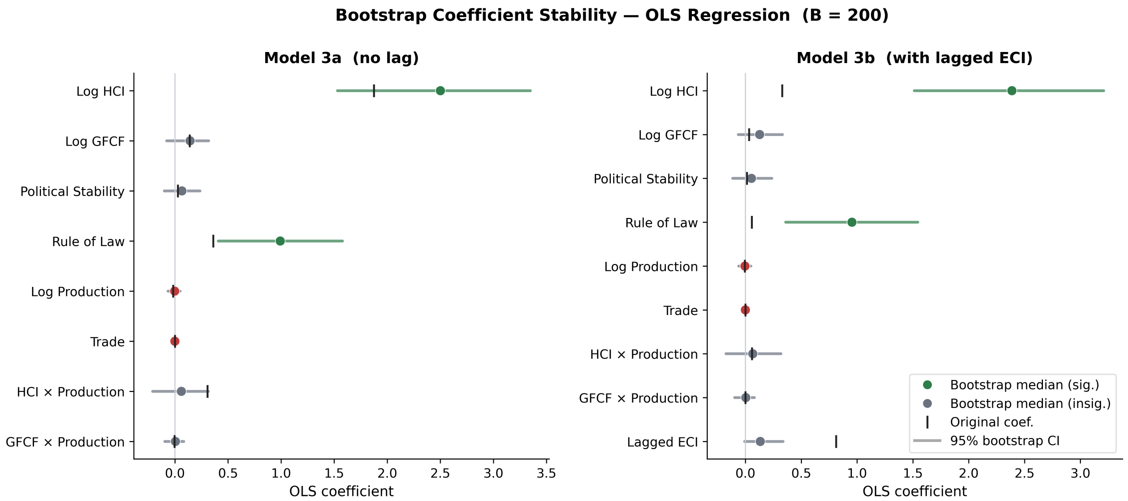

Figure F.1. OLS Regression: bootstrap coefficient confidence intervals

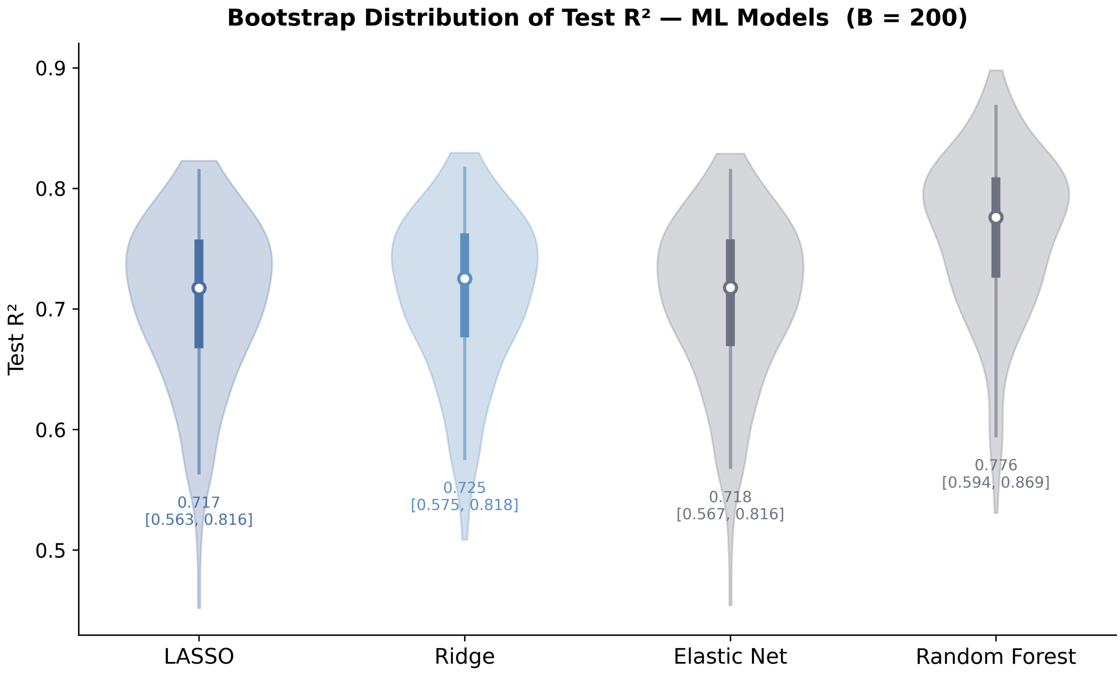

Figure F.2. ML Models: test R^2 distribution

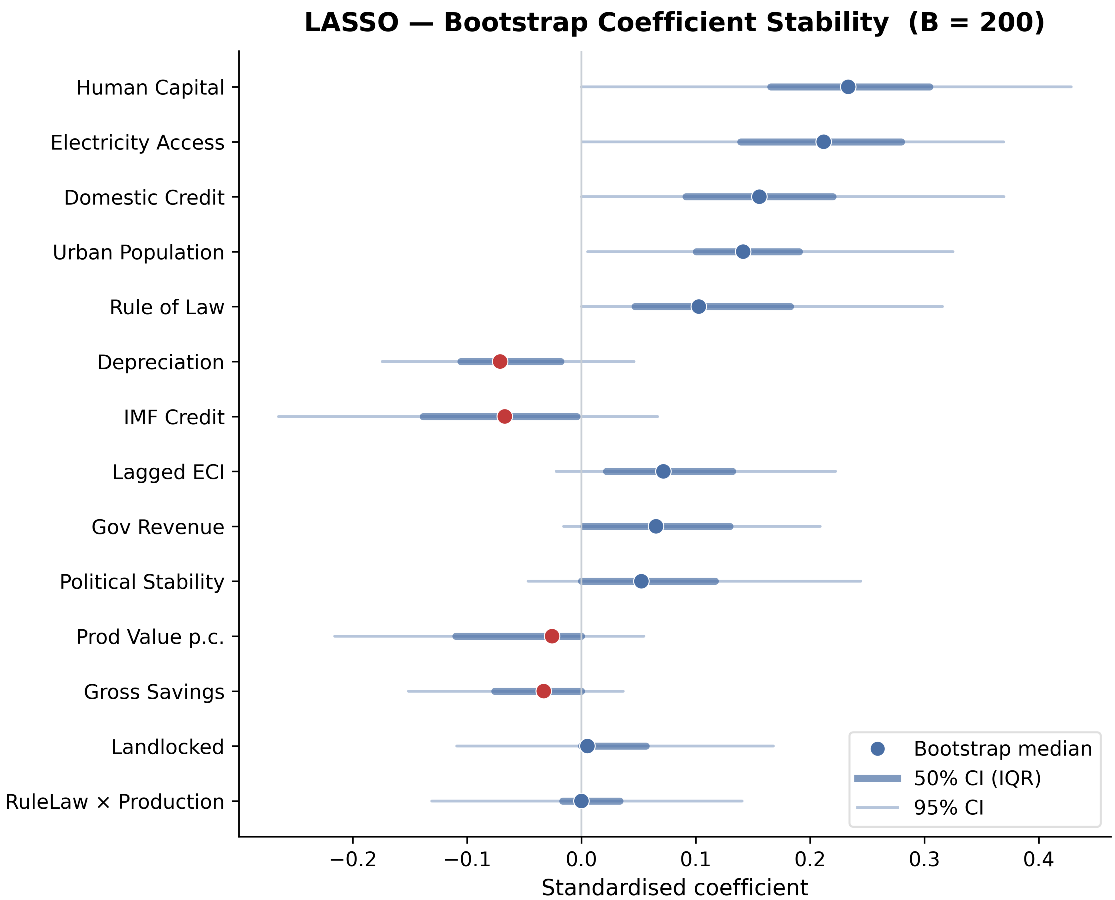

Figure F.3. LASSO: bootstrap coefficient stability

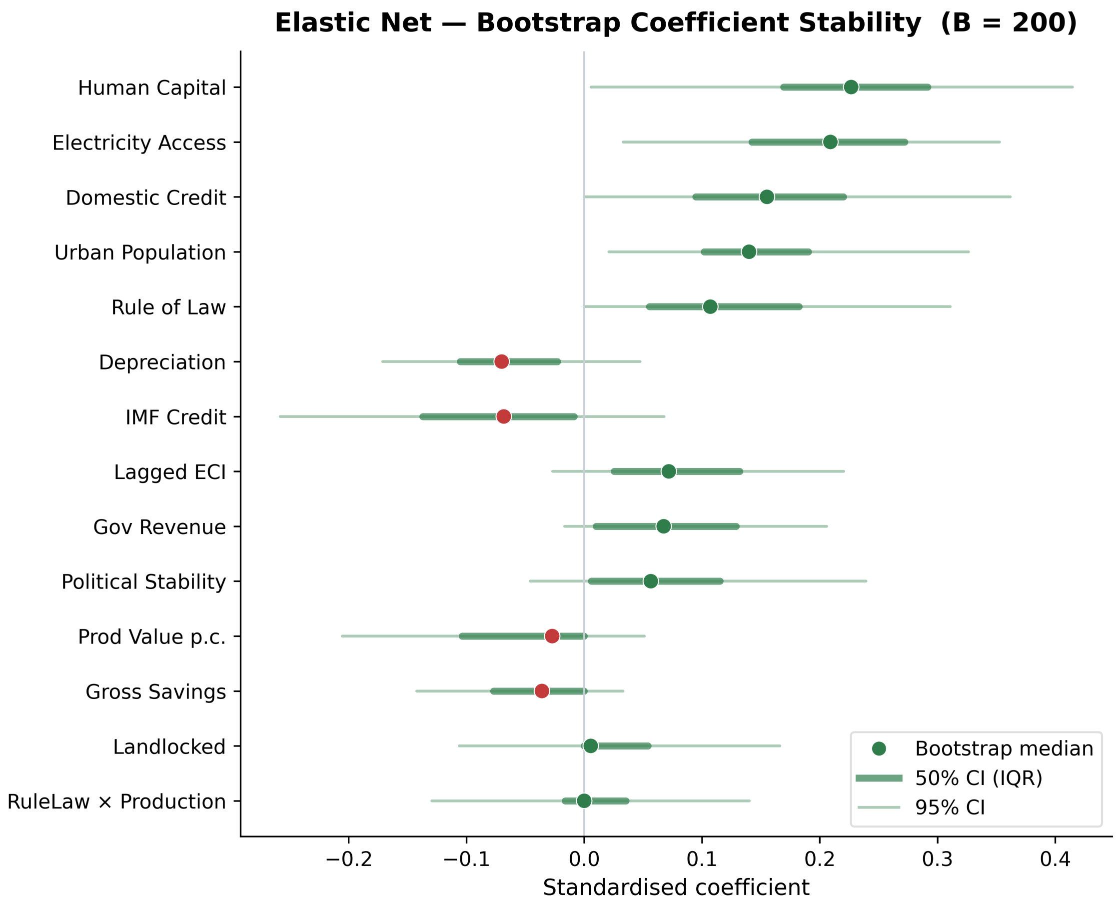

Figure F.4. Elastic Net: bootstrap coefficient stability

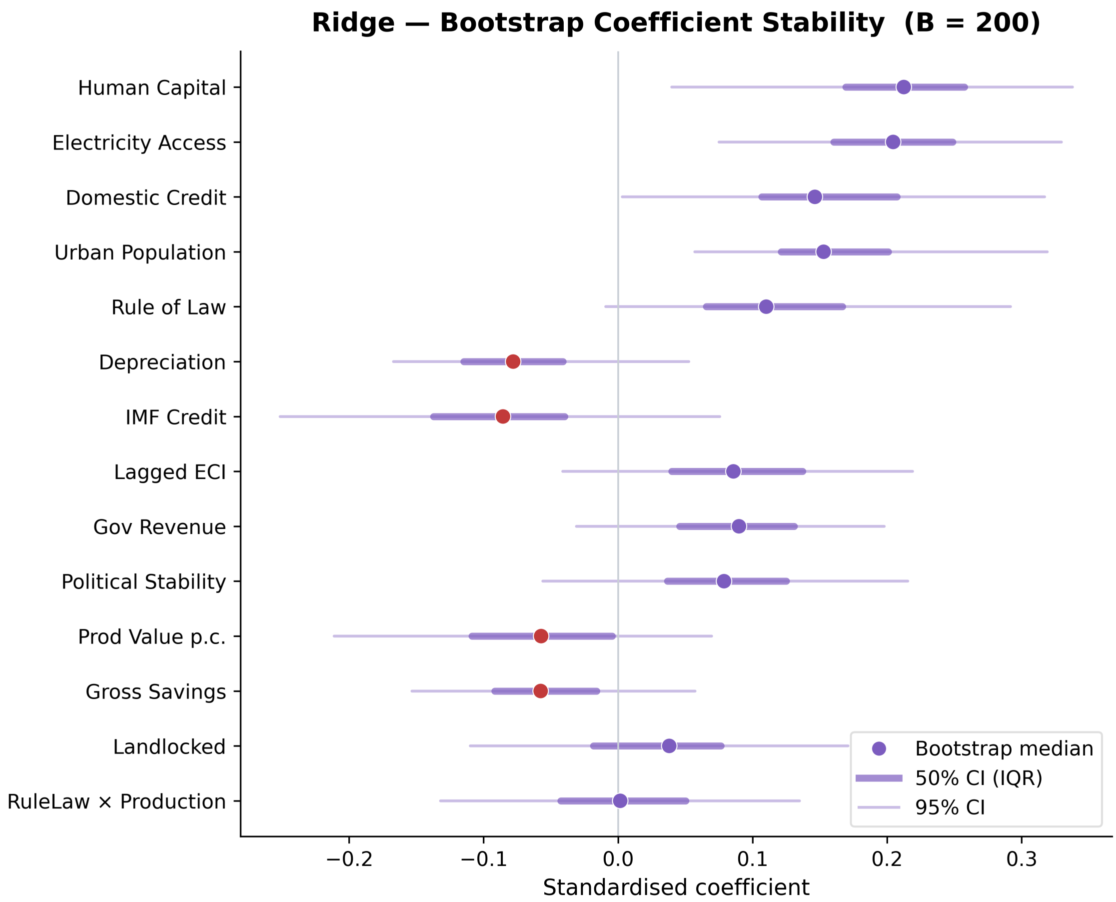

Figure F.5. Ridge: bootstrap coefficient stability

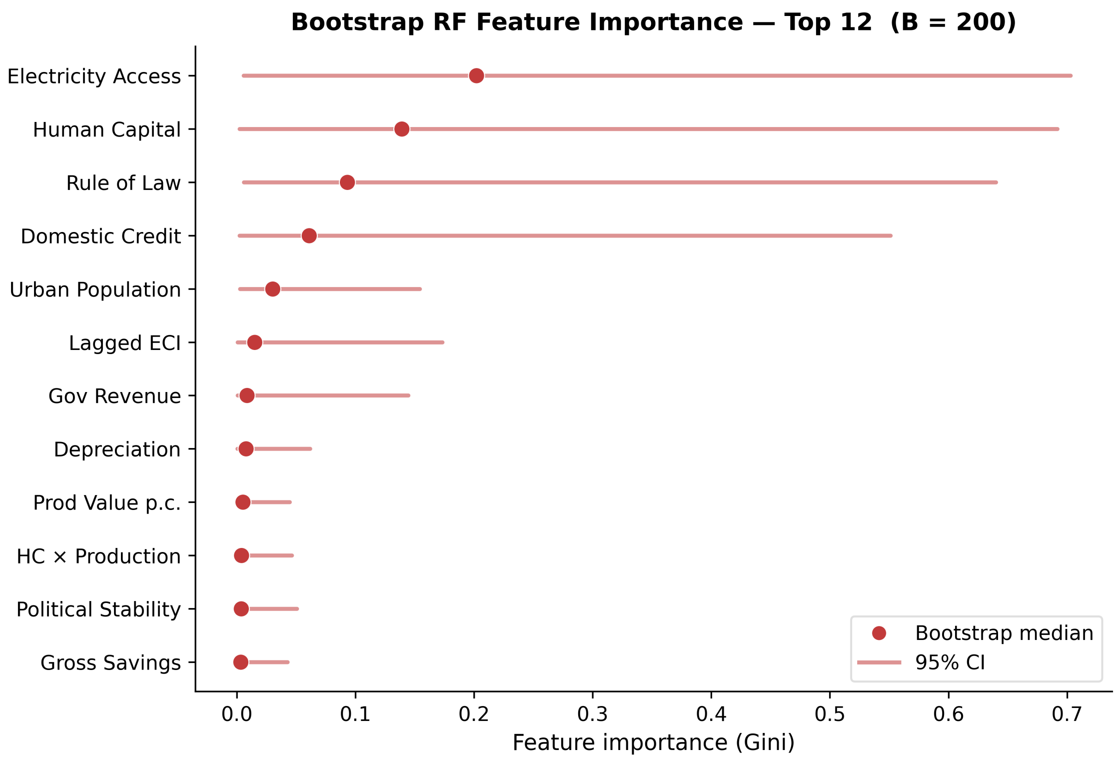

Figure F.6. Random Forest: bootstrap feature importance

F.2 Leave-One-Country-Out (LOCO) Analysis

To assess whether the econometric and machine learning results are driven by any single country, we implement a Leave-One-Country-Out (LOCO) analysis. For each of the 54 countries in the sample, we exclude all of its observations and refit both OLS specifications and all four ML models (LASSO, Ridge, Elastic Net, and Random Forest) on the remaining 53 countries, holding all hyperparameters and estimation procedures identical to the main specifications. This yields 54 iterations per model. We extract two diagnostics from each: coefficient stability, which tests whether the direction and magnitude of each predictor hold when any single country is removed, and R^2 stability, which tests whether explanatory power is distributed broadly across the sample or concentrated in a few observations.

Starting with the OLS models, the main predictors are highly stable. Log HCI, Rule of Law, and Political Stability are 100% sign-consistent across all runs in both models. Log GFCF is also stable in direction, retaining a positive sign across all exclusions. Two variables show weaker stability: Log Production Value, which changes sign in a small number of runs, and the GFCF $\times$ Production interaction, which shows weak sign instability in line with its near-zero full-sample estimate. R^2 tells the same story: the gap of approximately 0.43 between the two models is present for every single excluded country, confirming that the explanatory power contributed by lagged ECI is large.

The ML models show comparable stability. All three penalised models agree on sign for the top-ranked features across all 54 runs: Domestic Credit, HC $\times$ Production, and Electricity Access are consistently positive. Test R^2 is equally stable. The standard deviation of R^2 across the 54 iterations is 0.005 to 0.006 for every model, and no single country's exclusion shifts R^2 by more than approximately 0.04 absolute points from the full-sample baselines. Random Forest is consistently the highest-performing and most stable model across exclusions. Taken together, the LOCO results confirm that neither the OLS nor the ML findings are contingent on any single country's presence in the sample.

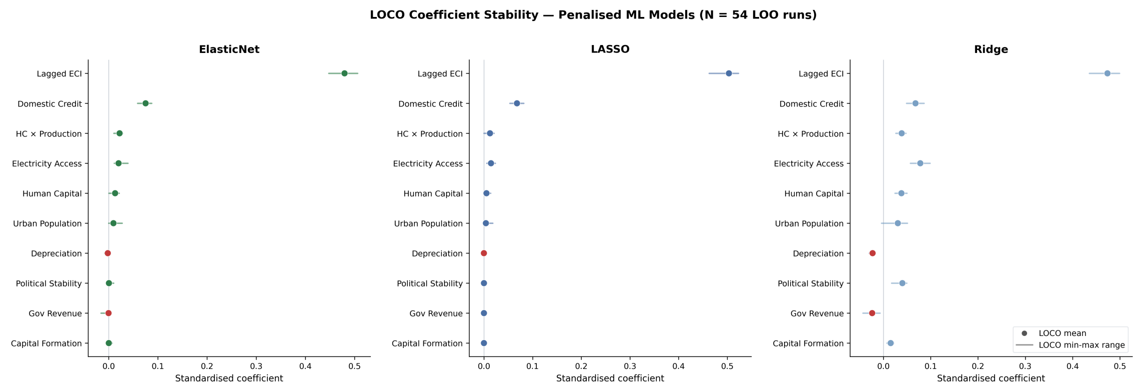

Figure F.7. LOCO coefficient stability: ML models

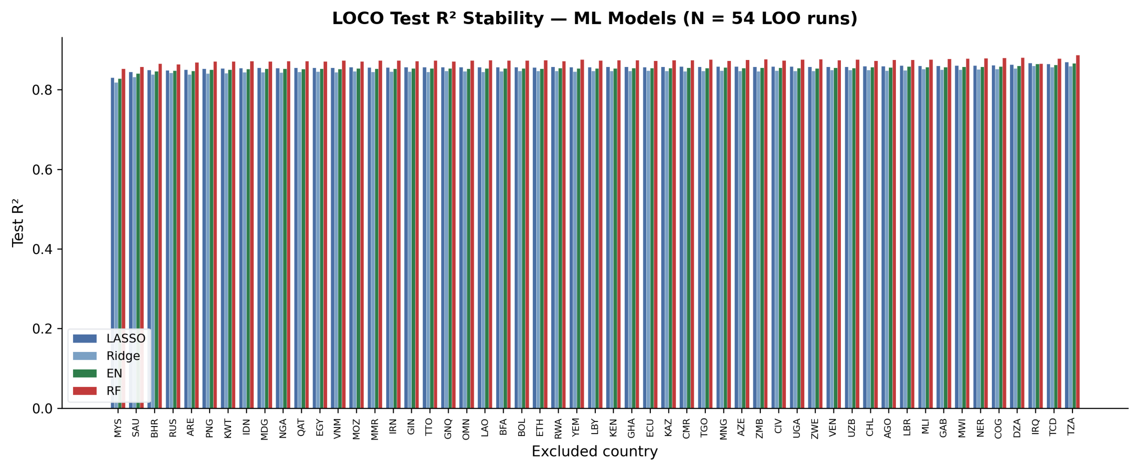

Figure F.8. LOCO R^2 stability: ML models

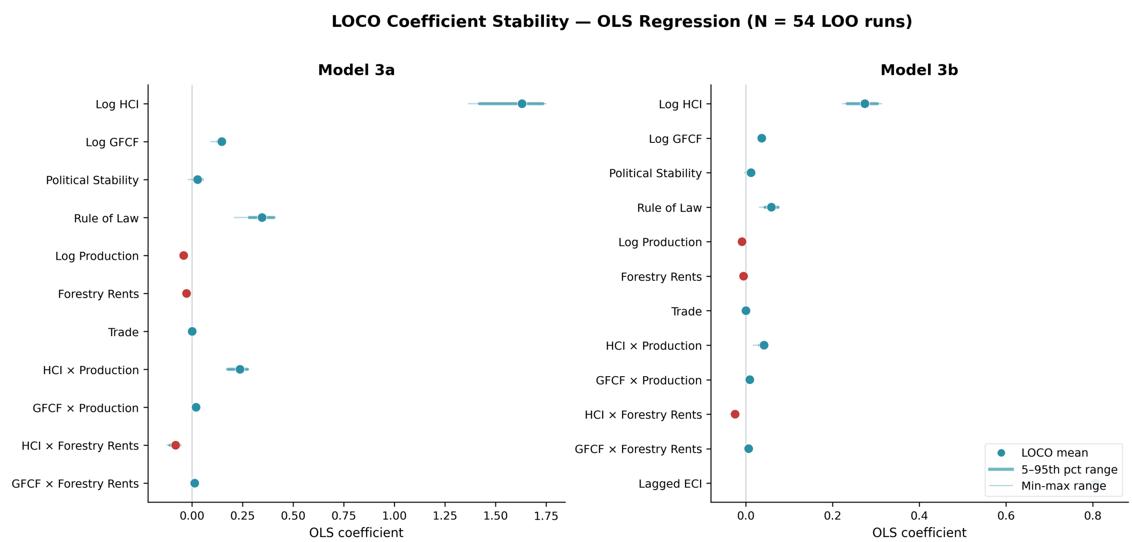

Figure F.9. LOCO coefficient stability: OLS models

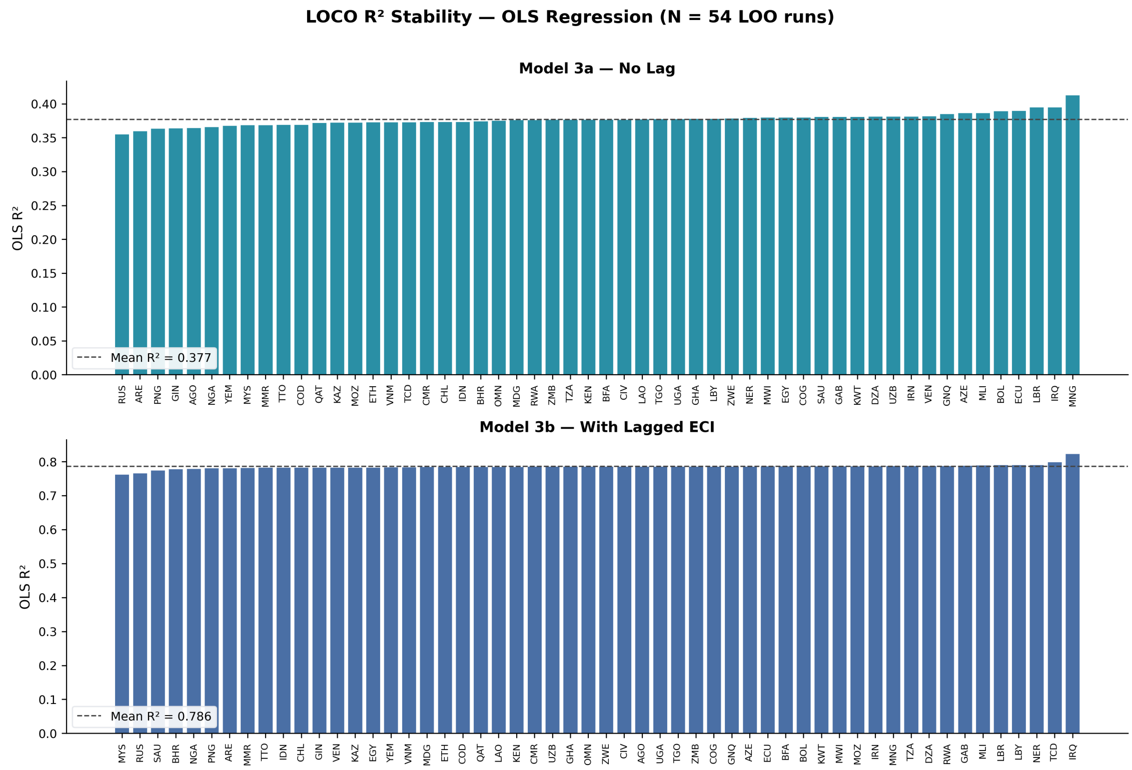

Figure F.10. LOCO R^2 stability: OLS models

Terms of Reference

Industrial Upgrading in Emerging Markets with Rich Natural Resources

Background

Understanding the drivers of economic growth is central to development research and sovereign risk assessment. Countries with high-value-added industries tend to achieve stronger long-term growth, fiscal stability, and economic resilience. While natural resource wealth offers emerging markets opportunities for industrial upgrading, outcomes vary dramatically. This divergence is closely linked to sovereign creditworthiness.

Research Questions

The project will identify economic indicators, institutional factors, and policy conditions that support or hinder industrial upgrading in resource-rich emerging markets.

Primary Questions

How successful have emerging market economies been at developing value-added industries from natural resource extraction? What factors explain the divergence across countries?

Secondary Questions

1. How sustainable are these shifts in terms of fiscal revenues, employment and export diversification?

2. How have they supported or hindered credit quality?
3. What policy levers and institutional conditions support or hinder the transition?

Project Objectives

Main Objective

To develop a framework that identifies and analyses the economic and institutional drivers of industrial upgrading in resource-rich emerging markets, understanding how this is linked with credit risk.

Specific Objectives

- Identify cross-country patterns to evaluate the predictors of industrial upgrading using supervised and unsupervised machine learning techniques.
- Build econometric models, founded on economic theory and literature, to complement and validate machine learning approaches.
- Conduct comparative country case studies to build a deeper understanding of local factors that influence industrial upgrading.
- Generate actionable recommendations for sovereign risk assessments.
- Deliver an Insights Report and a Presentation suitable for publication.

Scope

This project focuses on emerging and developing economies, recognising their unique structural challenges and opportunities in resource-based development pathways. Rather than examining industrial growth broadly, the analysis specifically targets value-added industrial upgrading, the critical transition from raw material extraction to higher-value manufacturing and processing activities.

Research Methodology

Literature Review

The project draws on economic complexity and value-added upgrading (Hidalgo & Hausmann, 2009) combined with the Growth Diagnostics Framework (Hausmann, Rodrik & Velasco, 2005); Acemoglu & Robinson's (2012) institutional economics framework; and Global Value Chain literature.

Analytical Approach

- Unsupervised learning (PCA, Clustering) to identify country groupings.
- Supervised learning (Ridge, Elastic Net, LASSO, Random Forest) to identify predictors.
- Econometric panel data modelling to complement findings.
- Case studies to examine mechanisms, governance factors, and policy interventions.

Project Phases and Timeline

Phase 1: Exploratory Analysis (Oct–Dec 2025)

Kick-off meeting, literature review, data cleaning, cluster analysis. Deliverables: ToR and progress presentation.

Phase 2: Supervised Modelling and Econometrics (Dec 2025 – Jan 2026)

Build and test supervised ML models, identify key predictors, select countries for case studies, econometric analysis.

Phase 3: Case Studies and Synthesis (Feb–Mar 2026)

Conduct case studies, integrate quantitative and qualitative insights, draft final report. Final presentation to Moody's (late March 2026).

Final Deliverables

Comprehensive research report, Moody’s Insights report, presentation slides.